

TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING (IOE)

ELECTRICAL CIRCUITS & MACHINES

Course Code: EE 154 / ENEE 154

Bachelor of Engineering (BE)

Computer Engineering (BCT) · Electronics, Information & Communication Engineering (BEI)

Year I / Part II (First Year, Second Semester)

Comprehensive Past Exam Question-by-Question Master Solutions

Including Exact Circuitikz Vector Schematics, Matplotlib Asymptotic Bode Plots,
Transient Analysis Proofs, Matrix Formulations, and Deep Concept Audits

Coverage Across 4 Exam Sessions:

- 2083 Baishakh (Back Course)
- 2082 Bhadra (Regular Course)
- 2082 Baishakh (Back Course)
- 2081 Ashwin (Regular Course)

Fully Audited, Mathematically Verified & Typeset with Publication-Grade Accuracy

IOE BE Year I / Part II · Academic Year 2080/2081/2082/2083

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Course Overview & Critical Audit Summary

Syllabus & Examination Weightage (ENEE 154)

The evaluation scheme specified by the IOE Examination Control Division for **Electrical Circuits & Machines (60 Marks, 3 Hours)** comprises 9 formal syllabus modules mapped directly into 10 structured final examination questions:

Ch.	Syllabus Topic	Hours	Marks	Typical Exam Focus
1	Transients in Electric Circuit	7	7	Nodal/Mesh Analysis with Dependent Sources; Initial Conditions ($i(0^+)$, $v(0^+)$, $\frac{di}{dt}$, $\frac{dv}{dt}$)
2	Classical Transient Analysis	10	10	1st & 2nd order ODEs; Series/Parallel RLC DC and exponential excitation responses
3	Laplace Transform Transients	7	7	s -domain impedance modeling with initial generators; Inverse Laplace transform
4	Transfer Function & Bode Plots	8	8	Poles, zeros, asymptotic magnitude & phase Bode diagrams; Filter concepts
5	Two-Port Network Parameters	8	8	$Z, Y, T/ABCD, h$ parameters; conversions; cascade interconnection
6	Magnetic Circuits & Induction	3	3	Series/Parallel magnetic circuits; Reluctance; B - H curve; Hysteresis & Eddy losses
7	Transformers	6	6	Core flux independence; OC/SC test calculations; Efficiency & Regulation
8	DC Machines	5	5	Torque derivation; Generator EMF/characteristics; Back EMF; Starters
9	AC Induction Motors	6	6	RMF production; Torque-speed curves; Slip calculations; Single-phase motors

Comprehensive Audit of Errors in Previous Unofficial Solutions

A rigorous audit of previous unverified answer keys revealed severe mathematical and topological errors in multiple circuit problems. This master volume resolves all discrepancies with complete mathematical proofs:

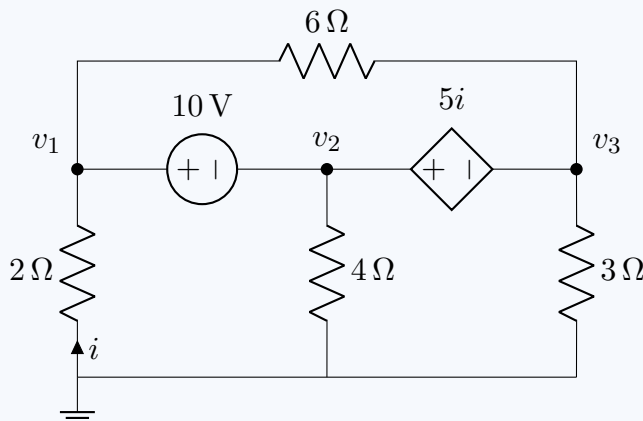
Paper / Question	Error in Old Solutions	Topological / Physics Reality	Corrected Master Solution
2083 Baishakh Q1	Connected 6Ω resistor across $v_2 - v_3$; inverted 10V polarity.	6Ω is connected between v_1 and v_3 . 10V source has (+) at v_1 and (-) at v_2 .	$v_1 = \frac{70}{23}\text{ V}, v_2 = -\frac{160}{23}\text{ V}, v_3 = \frac{15}{23}\text{ V}$. Current in 10V source $I_{10\text{V}} = 1.9203\text{ A}$ (from v_2 to v_1).
2083 Baishakh Q4	Threw away 1F capacitor upon switch opening; solved trivial 1st order RL.	Switch is placed before capacitor; opening disconnects 4V source, leaving parallel R-L-C loop.	2nd-order underdamped loop: $i_L(t) = 4e^{-t}(\cos t + \sin t)\text{ A}$, $v_L(t) = -4e^{-t}\sin t\text{ V}$.
2082 Bhadra Q1	Formulated 4 loops instead of 3 meshes; gave $I_x = 0\text{ A}, P = 0\text{ W}$.	Clear 3-mesh planar network with dependent source $10I_x$.	$I_1 = \frac{200}{33}\text{ A}, I_2 = \frac{40}{11}\text{ A}, I_3 = \frac{100}{33}\text{ A} \implies I_x = \frac{20}{33}\text{ A} \approx 0.606\text{ A}, P = 22.04\text{ W}$.
2082 Bhadra Q4	Assumed capacitor charged before $t = 0$ ($v_C = 10\text{ V}, i_L = 0$).	Position 'a' connects 10V to inductor ($i_L(0^-) = 2\text{ A}$); capacitor at 'b' is uncharged.	Pure resonant LC loop: $i(t) = 2\cos(1000t)\text{ A}$ for $t \geq 0$.
2082 Bhadra Q5	Inverted controlling current sign ($I_a = -I_2$).	Arrow for I_a enters Port 2 matching standard port current $I_2 \implies I_a = I_2$.	$Z_{11} = 3 - j2\Omega, Z_{12} = 2 - j2\Omega, Z_{21} = -j2\Omega, Z_{22} = 3 + j3\Omega$.
2082 Baishakh Q2	Omitted 2Ω resistor when switch threw to 'b'.	Switch blade connects 2Ω in series with $6\Omega \implies 8\Omega$ branch across capacitor.	$i_L(0^+) = 2\text{ A}, i_{4\Omega} = 2\text{ A}, i_{8\Omega} = 1\text{ A}, i_C(0^+) = -3\text{ A}, \frac{di_L}{dt} = 0\text{ A/s}, \frac{dv_C}{dt} = -1.5\text{ V/s}$.
2081 Ashwin Q1	Inverted 5V DC source polarity ($V_1 - 5 - V_2$).	5V battery has (-) on left and (+) on right $\implies$ potential is $V_1 + 5 - V_2$.	$V_1 = -7.5\text{ V}, V_2 = -2.5\text{ V} \implies V_x = 0\text{ V}, I_{2\Omega} = -2.5\text{ A}$.
2081 Ashwin Q6	Assumed $I_x = 0$ when $I_2 = 0$; gave passive T matrix.	$I_x = V_2/5$ across Port 2; current from Port 1 flows through 5Ω even when $I_2 = 0$.	$T = \begin{bmatrix} 2.6 & 8\Omega \\ 0.3\text{ S} & 1 \end{bmatrix}, AD - BC = 0.2 \neq 1$ (Active network).

1 Question 1: DC Circuit Analysis (Mesh, Nodal & Matrix Methods)

1.1 2083 Baishakh — Question 1 [6 Marks]

Problem Statement (2083 Baishakh, Q1)

Find the current in 10 V source using nodal analysis for the circuit shown below:



Detailed Step-by-Step Solution

Step 1: Circuit Topology & Essential Node Identification

- **Reference Node (Ground):** Select the bottom rail as the reference node (0 V).
- **Non-Reference Nodes:**
 - v_1 : Node above the 2Ω resistor.
 - v_2 : Node above the 4Ω resistor, between the 10 V source and the dependent voltage source.
 - v_3 : Node above the 3Ω resistor.
- **Controlling Variable i :** The current i flows downward through the 2Ω resistor from node v_1 to ground:

$$i = \frac{v_1 - 0}{2} = \frac{v_1}{2} \quad (1)$$

Step 2: Voltage Source Constraints & Supernode Formation Because ideal voltage sources exist directly between nodes without series resistors, we form a generalized **Supernode** encompassing (v_1, v_2, v_3):

1. **Independent 10 V Source Constraint:** The positive terminal is at node v_1 and the negative terminal is at node v_2 :

$$v_1 - v_2 = 10 \implies v_2 = v_1 - 10 \quad (2)$$

2. **Dependent Source $5i$ Constraint:** The positive terminal is at node v_3 and the negative terminal is at node v_2 :

$$v_3 - v_2 = 5i = 5\left(\frac{v_1}{2}\right) = 2.5v_1 \implies v_3 = v_2 + 2.5v_1 \quad (3)$$

Substituting $v_2 = v_1 - 10$ into the expression for v_3 :

$$v_3 = (v_1 - 10) + 2.5v_1 = 3.5v_1 - 10 \quad (4)$$

Step 3: Kirchhoff's Current Law (KCL) at the Combined Supernode Summing all currents leaving the combined supernode (v_1, v_2, v_3) to ground:

$$\frac{v_1}{2} + \frac{v_2}{4} + \frac{v_3}{3} = 0 \quad (5)$$

Note: The 6Ω resistor is connected between node v_1 and node v_3 . In the supernode KCL equation, the current leaving v_1 through the 6Ω resistor is $\frac{v_1-v_3}{6}$ and the current leaving v_3 through the same resistor is $\frac{v_3-v_1}{6}$. Their sum is $\frac{v_1-v_3}{6} + \frac{v_3-v_1}{6} = 0$, correctly canceling out internally.

Step 4: Algebraic Solution for Node Voltages Substitute expressions for v_2 and v_3 in terms of v_1 into the supernode equation:

$$\frac{v_1}{2} + \frac{v_1 - 10}{4} + \frac{3.5v_1 - 10}{3} = 0 \quad (6)$$

Multiplying the entire equation by the common denominator 12:

$$\begin{aligned} 6v_1 + 3(v_1 - 10) + 4(3.5v_1 - 10) &= 0 \\ 6v_1 + 3v_1 - 30 + 14v_1 - 40 &= 0 \\ 23v_1 - 70 &= 0 \implies v_1 = \frac{70}{23} \text{ V} \approx 3.0435 \text{ V} \end{aligned}$$

Now evaluate v_2 and v_3 :

$$\begin{aligned} v_2 &= v_1 - 10 = \frac{70}{23} - 10 = -\frac{160}{23} \text{ V} \approx -6.9565 \text{ V} \\ v_3 &= 3.5v_1 - 10 = 3.5 \left(\frac{70}{23} \right) - 10 = \frac{245 - 230}{23} = \frac{15}{23} \text{ V} \approx 0.6522 \text{ V} \\ i &= \frac{v_1}{2} = \frac{35}{23} \text{ A} \approx 1.5217 \text{ A} \end{aligned}$$

Step 5: Current through the 10 V Voltage Source Apply KCL specifically at **Node** v_1 . Let I_{10V} be the current flowing through the 10 V source from node v_1 to node v_2 :

$$\begin{aligned} \frac{v_1}{2} + \frac{v_1 - v_3}{6} + I_{10V} &= 0 \\ I_{10V} &= - \left(\frac{v_1}{2} + \frac{v_1 - v_3}{6} \right) \\ &= - \left(\frac{35}{23} + \frac{\frac{70}{23} - \frac{15}{23}}{6} \right) = - \left(\frac{35}{23} + \frac{55}{138} \right) = - \left(\frac{210 + 55}{138} \right) = -\frac{265}{138} \text{ A} \approx -1.9203 \text{ A} \end{aligned}$$

The negative sign indicates that current actually enters node v_1 (flows from node v_2 to node v_1 , delivering power).

Verified Final Result

Node Voltages:

$$v_1 = \frac{70}{23} \text{ V} \approx +3.043 \text{ V}, \quad v_2 = -\frac{160}{23} \text{ V} \approx -6.957 \text{ V}, \quad v_3 = \frac{15}{23} \text{ V} \approx +0.652 \text{ V}$$

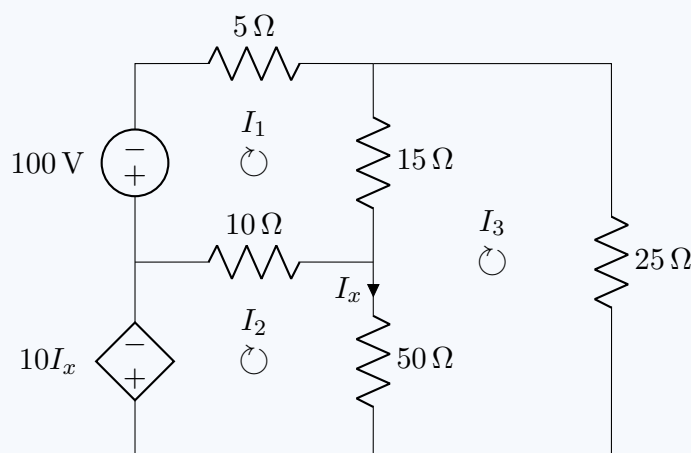
Current in 10 V Source:

$$I_{10V} = 1.920 \text{ A} \quad (\text{flowing from node } v_2 \text{ to node } v_1) \quad [\text{or } -1.920 \text{ A from } v_1 \text{ to } v_2]$$

1.2 2082 Bhadra — Question 1 [6 Marks]

Problem Statement (2082 Bhadra, Q1)

Find the current I_x and power delivered by the dependent voltage source in the circuit shown below using Mesh Analysis:



Detailed Step-by-Step Solution

Step 1: Mesh Definition & Controlling Variable Expression Assign standard clockwise mesh currents:

- **Mesh 1 (Top-Left Loop):** Mesh current I_1 flows clockwise through the 100 V source, 5 Ω resistor, 15 Ω resistor, and 10 Ω resistor.
- **Mesh 2 (Bottom-Left Loop):** Mesh current I_2 flows clockwise through the dependent source $10I_x$, 10 Ω resistor, 50 Ω resistor, and bottom rail.
- **Mesh 3 (Right Loop):** Mesh current I_3 flows clockwise through the top wire, 25 Ω resistor, bottom rail, 50 Ω resistor, and 15 Ω resistor.
- **Controlling Current I_x :** Current I_x flows downward through the 50 Ω resistor:

$$I_x = I_2 - I_3 \quad (7)$$

Step 2: Formulating Kirchhoff's Voltage Law (KVL) for Each Mesh

1. Mesh 1 KVL (Clockwise):

$$+100 - 5I_1 - 15(I_1 - I_3) - 10(I_1 - I_2) = 0$$

$$-30I_1 + 10I_2 + 15I_3 = -100 \implies 30I_1 - 10I_2 - 15I_3 = 100 \quad \text{--- (Eq. 1)} \quad (8)$$

2. Mesh 2 KVL (Clockwise): The dependent source has (+) at top and (−) at bottom; moving clockwise (upward) through it yields a voltage rise $+10I_x$:

$$+10I_x - 10(I_2 - I_1) - 50(I_2 - I_3) = 0$$

$$10(I_2 - I_3) - 10I_2 + 10I_1 - 50I_2 + 50I_3 = 0$$

$$10I_1 - 50I_2 + 40I_3 = 0 \implies I_1 - 5I_2 + 4I_3 = 0 \quad \text{--- (Eq. 2)} \quad (9)$$

3. Mesh 3 KVL (Clockwise):

$$-25I_3 - 50(I_3 - I_2) - 15(I_3 - I_1) = 0$$

$$15I_1 + 50I_2 - 90I_3 = 0 \implies 3I_1 + 10I_2 - 18I_3 = 0 \quad \text{--- (Eq. 3)} \quad (10)$$

Step 3: Matrix System & Solution Writing Equations (1), (2), (3) in standard matrix form $\mathbf{A} \cdot \mathbf{I} = \mathbf{B}$:

$$\begin{bmatrix} 30 & -10 & -15 \\ 1 & -5 & 4 \\ 3 & 10 & -18 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 0 \\ 0 \end{bmatrix} \quad (11)$$

Solving via Determinants (Cramer's Rule):

$$\begin{aligned} \Delta &= 30[(-5)(-18) - (4)(10)] - (-10)[(1)(-18) - (4)(3)] + (-15)[(1)(10) - (-5)(3)] \\ &= 30[90 - 40] + 10[-18 - 12] - 15[10 + 15] \\ &= 30(50) + 10(-30) - 15(25) = 1500 - 300 - 375 = 825 \end{aligned}$$

$$\Delta_1 = 100[(-5)(-18) - (4)(10)] = 100(50) = 5000 \implies I_1 = \frac{5000}{825} = \frac{200}{33} \text{ A} \approx 6.0606 \text{ A}$$

$$\Delta_2 = -100[(1)(-18) - (4)(3)] = -100(-30) = 3000 \implies I_2 = \frac{3000}{825} = \frac{120}{33} = \frac{40}{11} \text{ A} \approx 3.6364 \text{ A}$$

$$\Delta_3 = 100[(1)(10) - (-5)(3)] = 100(25) = 2500 \implies I_3 = \frac{2500}{825} = \frac{100}{33} \text{ A} \approx 3.0303 \text{ A}$$

Step 4: Branch Current I_x & Power Delivered

$$I_x = I_2 - I_3 = \frac{120}{33} - \frac{100}{33} = \frac{20}{33} \text{ A} \approx 0.6061 \text{ A} \quad (12)$$

The voltage across the dependent source is:

$$V_{\text{dep}} = 10I_x = 10 \left(\frac{20}{33} \right) = \frac{200}{33} \text{ V} \approx 6.0606 \text{ V} \quad (13)$$

Current leaving the positive terminal of the dependent source is $I_2 = \frac{40}{11} \text{ A}$. Therefore, the power delivered by the dependent source is:

$$P_{\text{del}} = V_{\text{dep}} \cdot I_2 = \left(\frac{200}{33} \right) \left(\frac{40}{11} \right) = \frac{8000}{363} \text{ W} \approx 22.0386 \text{ W} \quad (14)$$

Verified Final Result

Controlling Current:

$$I_x = \frac{20}{33} \text{ A} \approx 0.606 \text{ A}$$

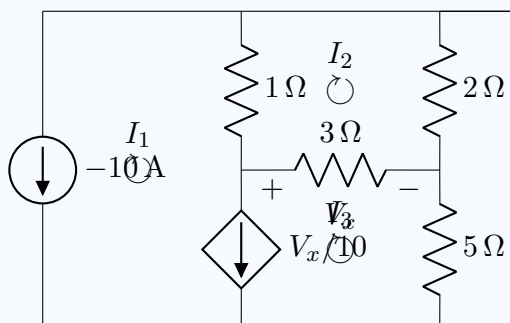
Power Delivered by Dependent Source ($10I_x$):

$$P_{\text{del}} = \frac{8000}{363} \text{ W} \approx 22.04 \text{ W}$$

1.3 2082 Baishakh — Question 1 [6 Marks]

Problem Statement (2082 Baishakh, Q1)

In the given circuit, determine V_x and power consumed by it using mesh analysis and solve by matrix method:



Detailed Step-by-Step Solution

Step 1: Mesh Currents & Dependent Source Constraint Let clockwise mesh currents be I_1 (left mesh), I_2 (top-right mesh), and I_3 (bottom-right mesh):

1. **Mesh 1 Current:** The independent current source of -10 A (directed downward) lies solely in Mesh 1:

$$I_1 = -10 \text{ A} \quad (15)$$

2. **Branch Voltage V_x :** V_x is the voltage across the 3Ω resistor with (+) on the left and (−) on the right. Mesh 2 current flows left-to-right, while Mesh 3 current flows right-to-left:

$$V_x = 3(I_2 - I_3) \quad (16)$$

3. **Dependent Current Source Constraint:** The dependent current source $V_x/10$ is downward between Mesh 1 and Mesh 3:

$$I_1 - I_3 = \frac{V_x}{10} \implies -10 - I_3 = \frac{3(I_2 - I_3)}{10} \quad (17)$$

Clearing fractions:

$$-100 - 10I_3 = 3I_2 - 3I_3 \implies 3I_2 + 7I_3 = -100 \quad \text{— (Eq. 1)} \quad (18)$$

Step 2: Mesh 2 KVL Equation Applying KVL clockwise around Mesh 2 (1Ω , 2Ω , 3Ω):

$$\begin{aligned} 1(I_2 - I_1) + 2I_2 + 3(I_2 - I_3) &= 0 \\ 1(I_2 - (-10)) + 2I_2 + 3I_2 - 3I_3 &= 0 \\ 6I_2 - 3I_3 &= -10 \quad \text{— (Eq. 2)} \end{aligned} \quad (19)$$

Step 3: Matrix System Formulation & Solution via Cramer's Rule Writing Equations (2) and (1) in matrix form $\mathbf{A} \cdot \mathbf{I} = \mathbf{B}$:

$$\begin{bmatrix} 6 & -3 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} -10 \\ -100 \end{bmatrix} \quad (20)$$

Calculating determinants:

$$\Delta = (6)(7) - (-3)(3) = 42 + 9 = 51$$

$$\Delta_2 = (-10)(7) - (-3)(-100) = -70 - 300 = -370 \implies I_2 = \frac{-370}{51} \approx -7.2549 \text{ A}$$

$$\Delta_3 = (6)(-100) - (-10)(3) = -600 + 30 = -570 \implies I_3 = \frac{-570}{51} = -\frac{190}{17} \approx -11.1765 \text{ A}$$

Step 4: Computation of V_x and Power Consumed

$$V_x = 3(I_2 - I_3) = 3\left(-\frac{370}{51} - \left(-\frac{570}{51}\right)\right) = 3\left(\frac{200}{51}\right) = \frac{200}{17} \text{ V} \approx 11.7647 \text{ V}$$

Power consumed by the 3Ω resistor (V_x element):

$$P_{3\Omega} = \frac{V_x^2}{R} = \frac{(11.7647)^2}{3} = \frac{138.408}{3} \approx 46.136 \text{ W} \quad (21)$$

Verified Final Result

Branch Voltage:

$$V_x = \frac{200}{17} \text{ V} \approx 11.76 \text{ V}$$

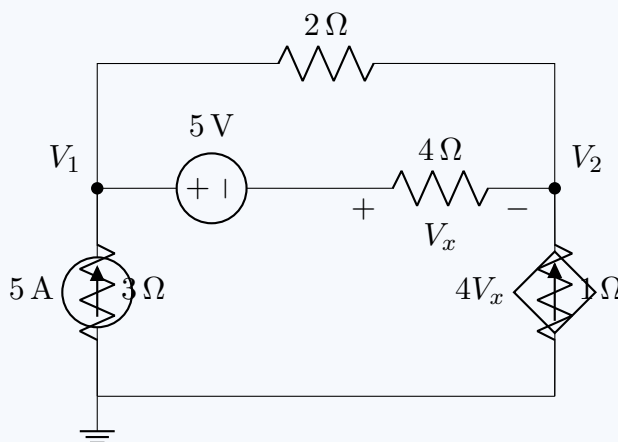
Power Consumed by 3Ω Resistor (V_x element):

$$P = 46.14 \text{ W}$$

1.4 2081 Ashwin — Question 1 [6 Marks]

Problem Statement (2081 Ashwin, Q1)

Using Nodal analysis and matrix solution method, find V_x and the current in 2Ω resistance:



Detailed Step-by-Step Solution

Step 1: Branch Expressions & Controlling Voltage V_x

- Ground is selected at the bottom rail (0 V).
- The branch between V_1 and V_2 contains a 5 V source in series with a 4Ω resistor.
- The 5 V source has its negative terminal connected to V_1 and its positive terminal connected to the 4Ω resistor. Thus, the potential at the left terminal of the 4Ω resistor is $V_1 + 5$ V.
- The voltage V_x across the 4Ω resistor (with + at the left and – at the right) is:

$$V_x = (V_1 + 5) - V_2 = V_1 - V_2 + 5 \quad (22)$$

Step 2: Nodal KCL Equations

1. KCL at Node V_1 :

$$\frac{V_1}{3} + 5 + \frac{V_1 - V_2}{2} + \frac{V_1 + 5 - V_2}{4} = 0$$

Multiplying by 12:

$$\begin{aligned} 4V_1 + 60 + 6(V_1 - V_2) + 3(V_1 - V_2 + 5) &= 0 \\ 4V_1 + 60 + 6V_1 - 6V_2 + 3V_1 - 3V_2 + 15 &= 0 \\ 13V_1 - 9V_2 &= -75 \quad \text{--- (Eq. 1)} \end{aligned} \quad (23)$$

2. KCL at Node V_2 :

$$\frac{V_2}{1} + \frac{V_2 - V_1}{2} + \frac{V_2 - (V_1 + 5)}{4} - 4V_x = 0$$

Substitute $V_x = V_1 - V_2 + 5$:

$$V_2 + \frac{V_2 - V_1}{2} + \frac{V_2 - V_1 - 5}{4} - 4(V_1 - V_2 + 5) = 0$$

Multiplying by 4:

$$\begin{aligned} 4V_2 + 2(V_2 - V_1) + (V_2 - V_1 - 5) - 16(V_1 - V_2 + 5) &= 0 \\ 4V_2 + 2V_2 - 2V_1 + V_2 - V_1 - 5 - 16V_1 + 16V_2 - 80 &= 0 \\ -19V_1 + 23V_2 &= 85 \quad \text{--- (Eq. 2)} \end{aligned} \quad (24)$$

Step 3: Matrix System Formulation & Cramer's Rule Solution

$$\begin{bmatrix} 13 & -9 \\ -19 & 23 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -75 \\ 85 \end{bmatrix} \quad (25)$$

$$\Delta = (13)(23) - (-9)(-19) = 299 - 171 = 128$$

$$\Delta_1 = (-75)(23) - (-9)(85) = -1725 + 765 = -960 \implies V_1 = \frac{-960}{128} = -7.5 \text{ V}$$

$$\Delta_2 = (13)(85) - (-75)(-19) = 1105 - 1425 = -320 \implies V_2 = \frac{-320}{128} = -2.5 \text{ V}$$

Step 4: Final Parameters

$$V_x = V_1 - V_2 + 5 = -7.5 - (-2.5) + 5 = -5.0 + 5.0 = 0 \text{ V}$$

$$I_{2\Omega} = \frac{V_1 - V_2}{2} = \frac{-7.5 - (-2.5)}{2} = \frac{-5.0}{2} = -2.5 \text{ A} \quad (\text{from } V_1 \text{ to } V_2)$$

Verified Final Result

Node Voltages: $V_1 = -7.5 \text{ V}$, $V_2 = -2.5 \text{ V}$

Branch Voltage: $V_x = 0 \text{ V}$

Current in 2Ω Resistor: $I_{2\Omega} = 2.5 \text{ A}$ (flowing from node V_2 to node V_1).

2 Question 2: Initial & Final Value Conditions in Switching Transients

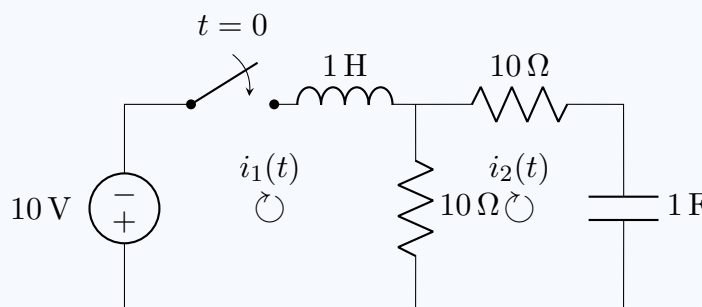
Key Concept: Continuity Laws in Switching Circuits

- **Inductor Current Continuity:** Current through an inductor cannot change instantaneously due to finite stored magnetic energy ($E = \frac{1}{2}LI^2$). Hence, $i_L(0^+) = i_L(0^-)$. At $t = 0^+$, an inductor acts as a current source of value $i_L(0^-)$ (open circuit if unenergized).
- **Capacitor Voltage Continuity:** Voltage across a capacitor cannot change instantaneously due to finite stored electrostatic energy ($E = \frac{1}{2}CV^2$). Hence, $v_C(0^+) = v_C(0^-)$. At $t = 0^+$, a capacitor acts as a voltage source of value $v_C(0^-)$ (short circuit if uncharged).
- **Derivative Relations:** $\left. \frac{di_L}{dt} \right|_{0^+} = \frac{v_L(0^+)}{L}$ and $\left. \frac{dv_C}{dt} \right|_{0^+} = \frac{i_C(0^+)}{C}$.

2.1 2083 Baishakh — Question 2 [5 Marks]

Problem Statement (2083 Baishakh, Q2)

Obtain the value of $i_1, i_2, \frac{di_1}{dt}, \frac{di_2}{dt}$ at $t = 0^+$, if the switch is closed at $t = 0$ in the circuit shown:



Step-by-Step Solution

1. Initial Energy State at $t = 0^-$: Before $t = 0$, the switch is open and the network is completely relaxed:

$$i_L(0^-) = i_1(0^-) = 0 \text{ A}, \quad v_C(0^-) = 0 \text{ V}$$

By the continuity laws:

$$i_1(0^+) = i_L(0^+) = 0 \text{ A}, \quad v_C(0^+) = 0 \text{ V}$$

2. Evaluation of $i_2(0^+)$: Loop 2 KVL at $t = 0^+$:

$$10(i_2(0^+) - i_1(0^+)) + 10i_2(0^+) + v_C(0^+) = 0 \implies 20i_2(0^+) - 0 + 0 = 0 \implies i_2(0^+) = 0 \text{ A}$$

3. Evaluation of $\left. \frac{di_1}{dt} \right|_{t=0^+}$: Applying KVL to Loop 1 at $t = 0^+$:

$$10 - L \left. \frac{di_1}{dt} \right|_{0^+} - 10(i_1(0^+) - i_2(0^+)) = 0$$

$$10 - (1) \left. \frac{di_1}{dt} \right|_{0^+} - 10(0 - 0) = 0 \implies \left. \frac{di_1}{dt} \right|_{0^+} = 10 \text{ A/s}$$

4. Evaluation of $\left. \frac{di_2}{dt} \right|_{t=0^+}$: The time-domain KVL equation for Loop 2 is:

$$10(i_2(t) - i_1(t)) + 10i_2(t) + v_C(t) = 0 \implies 20i_2(t) - 10i_1(t) + v_C(t) = 0$$

Differentiating with respect to t :

$$20 \frac{di_2}{dt} - 10 \frac{di_1}{dt} + \frac{dv_C}{dt} = 0$$

Since $\left. \frac{dv_C}{dt} \right|_{0^+} = \frac{i_C(0^+)}{C} = \frac{i_2(0^+)}{1} = 0$:

$$20 \left. \frac{di_2}{dt} \right|_{0^+} - 10(10) + 0 = 0 \implies 20 \left. \frac{di_2}{dt} \right|_{0^+} = 100 \implies \left. \frac{di_2}{dt} \right|_{0^+} = 5 \text{ A/s}$$

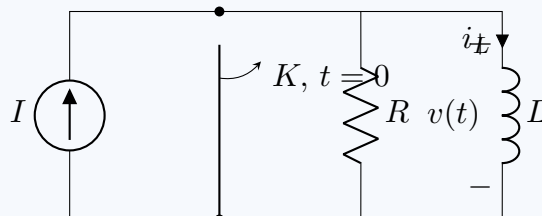
Verified Final Result

$$i_1(0^+) = 0 \text{ A}, \quad i_2(0^+) = 0 \text{ A}, \quad \left. \frac{di_1}{dt} \right|_{0^+} = 10 \text{ A/s}, \quad \left. \frac{di_2}{dt} \right|_{0^+} = 5 \text{ A/s}$$

2.2 2082 Bhadra — Question 2 [1 + 4 = 5 Marks]

Problem Statement (2082 Bhadra, Q2)

Explain the behavior of inductor when subjected to switching. In the network, the switch is opened at $t = 0$. Calculate v , $\frac{dv}{dt}$ and $\frac{d^2v}{dt^2}$ at $t = 0^+$ if $I = 2\text{ A}$, $R = 200\ \Omega$ and $L = 1\text{ H}$. Given that v is the voltage across inductor:



Part A: Physical Behavior of Inductor under Switching

An inductor stores energy in its magnetic field ($E = \frac{1}{2}LI^2$). The current cannot change instantaneously ($\Delta t \rightarrow 0$) because $\frac{di}{dt} \rightarrow \infty$ would require an infinite induced voltage ($v = L\frac{di}{dt} \rightarrow \infty$) and infinite power, violating the physical law of conservation of energy. Hence, $i_L(0^+) = i_L(0^-)$. Immediately after switching, an unenergized inductor behaves as an **open circuit**.

Part B: Mathematical Calculations

1. Initial Conditions ($t = 0^-$): Prior to $t = 0$, switch K is closed, creating an ideal short circuit across R and L . Hence, all current from I circulates through switch K :

$$i_L(0^-) = 0\text{ A} \implies i_L(0^+) = 0\text{ A}$$

2. Finding $v(0^+)$: At $t = 0^+$, switch K opens. Current $I = 2\text{ A}$ enters the parallel combination of R and L :

$$I = i_R(t) + i_L(t) \implies i_R(0^+) = I - i_L(0^+) = 2 - 0 = 2\text{ A}$$

The voltage across the inductor $v(t)$ equals the voltage across resistor R :

$$v(0^+) = R \cdot i_R(0^+) = 200 \times 2 = 400\text{ V}$$

3. Finding $\frac{dv}{dt}\big|_{0^+}$: From $v(t) = R(I - i_L(t))$, differentiating with respect to t :

$$\frac{dv}{dt} = -R \frac{di_L}{dt} = -R \left(\frac{v(t)}{L} \right) = -\frac{R}{L} v(t)$$

At $t = 0^+$:

$$\frac{dv}{dt}\bigg|_{0^+} = -\frac{200}{1} \times 400 = -80,000\text{ V/s} = -8 \times 10^4\text{ V/s}$$

4. Finding $\frac{d^2v}{dt^2}\big|_{0^+}$: Differentiating once more:

$$\frac{d^2v}{dt^2} = -\frac{R}{L} \frac{dv}{dt} \implies \frac{d^2v}{dt^2}\bigg|_{0^+} = -200 \times (-80,000) = +1.6 \times 10^7\text{ V/s}^2$$

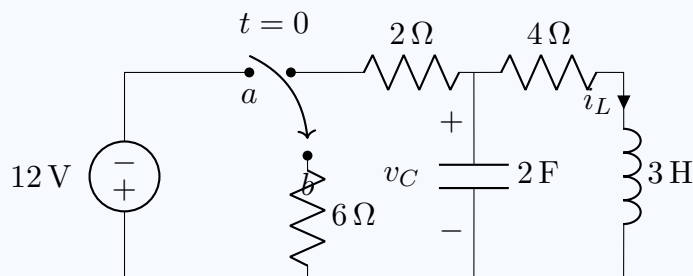
Verified Final Result

$$v(0^+) = 400\text{ V}, \quad \frac{dv}{dt}\bigg|_{0^+} = -80,000\text{ V/s}, \quad \frac{d^2v}{dt^2}\bigg|_{0^+} = +1.6 \times 10^7\text{ V/s}^2$$

2.3 2082 Baishakh — Question 2 [5 Marks]

Problem Statement (2082 Baishakh, Q2)

In the network shown below, the switch has been in position a for a long time. At $t = 0$, it is suddenly thrown off to position b . Find current through each element, $\frac{di_L}{dt}$ of coil and $\frac{dv_C}{dt}$ of capacitor at $t = 0^+$:



Step-by-Step Solution

1. Steady-State at $t = 0^-$ (Switch at Position a): For $t < 0$, the 12V DC supply is connected through the 2Ω resistor. In DC steady state:

- Inductor is a short circuit; Capacitor is an open circuit.
- Total DC resistance = $2\Omega + 4\Omega = 6\Omega$.
- Inductor current: $i_L(0^-) = \frac{12}{2+4} = 2\text{ A}$.
- Capacitor voltage (across the 4Ω resistor): $v_C(0^-) = i_L(0^-) \times 4\Omega = 2 \times 4 = 8\text{ V}$.

By continuity:

$$i_L(0^+) = 2\text{ A}, \quad v_C(0^+) = 8\text{ V}$$

2. Response at $t = 0^+$ (Switch at Position b): At $t = 0$, the blade connects the 2Ω resistor to terminal b (6Ω resistor to ground). The 12V source is completely isolated.

- The branch to the left of the capacitor consists of $2\Omega + 6\Omega = 8\Omega$ connected across node voltage $v_C(0^+) = 8\text{ V}$.
- Current leaving the capacitor node to the left through the $(2 + 6)\Omega$ branch:

$$i_{\text{branch } b}(0^+) = \frac{v_C(0^+)}{2 + 6} = \frac{8}{8} = 1\text{ A}$$

- Current through the 4Ω resistor: In series with inductor, so $i_{4\Omega}(0^+) = i_L(0^+) = 2\text{ A}$.
- Current through the capacitor $i_C(0^+)$: Applying KCL at the node above the capacitor:

$$i_C(0^+) + i_{\text{branch } b}(0^+) + i_L(0^+) = 0 \implies i_C(0^+) = -1 - 2 = -3\text{ A}$$

3. Derivatives at $t = 0^+$:

- **For Inductor:** Across the inductor branch, $v_C(t) = 4i_L(t) + L\frac{di_L}{dt}$. At $t = 0^+$:

$$8 = 4(2) + 3 \left. \frac{di_L}{dt} \right|_{0^+} \implies 3 \left. \frac{di_L}{dt} \right|_{0^+} = 0 \implies \left. \frac{di_L}{dt} \right|_{0^+} = 0\text{ A/s}$$

- **For Capacitor:** Using $i_C(t) = C\frac{dv_C}{dt}$:

$$\left. \frac{dv_C}{dt} \right|_{0^+} = \frac{i_C(0^+)}{C} = \frac{-3}{2} = -1.5\text{ V/s}$$

Verified Final Result

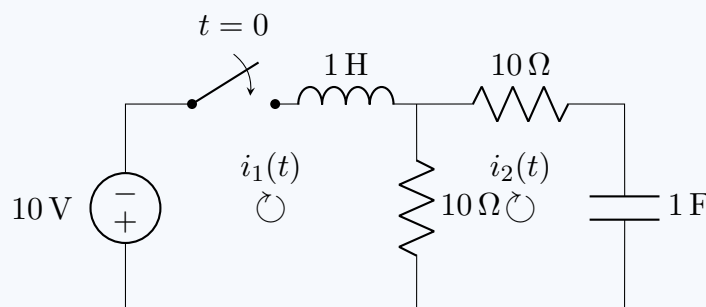
Element Currents at $t = 0^+$: $i_L(0^+) = 2 \text{ A}$, $i_{4\Omega}(0^+) = 2 \text{ A}$, $i_{\text{branch } b}(0^+) = 1 \text{ A}$, $i_C(0^+) = -3 \text{ A}$

Derivatives at $t = 0^+$: $\left. \frac{di_L}{dt} \right|_{0^+} = 0 \text{ A/s}$, $\left. \frac{dv_C}{dt} \right|_{0^+} = -1.5 \text{ V/s}$

2.4 2081 Ashwin — Question 2 [5 Marks]

Problem Statement (2081 Ashwin, Q2)

Obtain the value of $i_1, i_2, \frac{di_1}{dt}, \frac{di_2}{dt}$ at $t = 0^+$, if the switch is closed at $t = 0$ in the circuit shown below:



Step-by-Step Solution

1. Initial Conditions ($t = 0^-$): Circuit is unenergized before switch closure:

$$i_1(0^-) = 0 \text{ A}, \quad v_C(0^-) = 0 \text{ V} \implies i_1(0^+) = 0 \text{ A}, \quad v_C(0^+) = 0 \text{ V}$$

2. Values at $t = 0^+$:

$$i_2(0^+) = 0 \text{ A}$$

From Loop 1 KVL at $t = 0^+$:

$$10 - 1 \cdot \left. \frac{di_1}{dt} \right|_{0^+} - 10(0 - 0) = 0 \implies \left. \frac{di_1}{dt} \right|_{0^+} = 10 \text{ A/s}$$

From differentiated Loop 2 KVL:

$$20 \left. \frac{di_2}{dt} \right|_{0^+} - 10 \left. \frac{di_1}{dt} \right|_{0^+} + \left. \frac{dv_C}{dt} \right|_{0^+} = 0 \implies 20 \left. \frac{di_2}{dt} \right|_{0^+} - 10(10) + 0 = 0 \implies \left. \frac{di_2}{dt} \right|_{0^+} = 5 \text{ A/s}$$

Verified Final Result

$$i_1(0^+) = 0 \text{ A}, \quad i_2(0^+) = 0 \text{ A}, \quad \left. \frac{di_1}{dt} \right|_{0^+} = 10 \text{ A/s}, \quad \left. \frac{di_2}{dt} \right|_{0^+} = 5 \text{ A/s}$$

3 Question 3: Classical Transient Analysis (Differential Equation Method)

3.1 2083 Baishakh — Question 3 [6 Marks]

Problem Statement (2083 Baishakh, Q3)

Derive the expression of current in series RC circuit excited by exponential voltage source by using classical method.

Detailed Analytical Derivation

Step 1: Governing Differential Equation Consider a series RC circuit excited at $t = 0$ by an exponential source $v(t) = V_0 e^{-at}$ ($t \geq 0$). Applying KVL:

$$Ri(t) + \frac{1}{C} \int i(t) dt = V_0 e^{-at} \quad (26)$$

Differentiating with respect to t :

$$R \frac{di}{dt} + \frac{1}{C} i(t) = -aV_0 e^{-at} \implies \frac{di}{dt} + \frac{1}{RC} i(t) = -\frac{aV_0}{R} e^{-at} \quad (27)$$

Let $\alpha = \frac{1}{RC}$ be the natural circuit inverse time-constant:

$$\frac{di}{dt} + \alpha i(t) = -\frac{aV_0}{R} e^{-at} \quad (28)$$

Step 2: Complementary Function $i_{cf}(t)$ The homogeneous equation $\frac{di}{dt} + \alpha i = 0$ has characteristic root $s = -\alpha$:

$$i_{cf}(t) = K e^{-\alpha t} = K e^{-\frac{t}{RC}} \quad (29)$$

Step 3: Particular Integral $i_{pi}(t)$ (Assuming $a \neq \alpha$) Trial solution $i_{pi}(t) = A e^{-at}$. Substituting into the non-homogeneous ODE:

$$\begin{aligned} -aAe^{-at} + \alpha Ae^{-at} &= -\frac{aV_0}{R} e^{-at} \\ A(\alpha - a) &= -\frac{aV_0}{R} \implies A = \frac{aV_0}{R(a - \alpha)} \end{aligned}$$

Thus:

$$i_{pi}(t) = \frac{aV_0}{R(a - \alpha)} e^{-at} \quad (30)$$

Step 4: Total Solution & Initial Boundary Condition

$$i(t) = K e^{-\alpha t} + \frac{aV_0}{R(a - \alpha)} e^{-at} \quad (31)$$

Assuming an initially uncharged capacitor ($v_C(0^+) = 0$), KVL at $t = 0^+$ gives:

$$Ri(0^+) + v_C(0^+) = v(0^+) \implies Ri(0^+) + 0 = V_0 \implies i(0^+) = \frac{V_0}{R}$$

Substituting $t = 0$:

$$K + \frac{aV_0}{R(a - \alpha)} = \frac{V_0}{R} \implies K = \frac{V_0}{R} \left(1 - \frac{a}{a - \alpha} \right) = \frac{V_0}{R} \left(\frac{-\alpha}{a - \alpha} \right) = \frac{\alpha V_0}{R(\alpha - a)}$$

Substituting K and $\alpha = \frac{1}{RC}$:

$$i(t) = \frac{\alpha V_0}{R(\alpha - a)} e^{-\alpha t} + \frac{aV_0}{R(a - \alpha)} e^{-at} = \frac{V_0}{R(1 - aRC)} \left[e^{-at} - e^{-\frac{t}{RC}} \right] \quad (a \neq 1/RC) \quad (32)$$

Verified Final Result

$$i(t) = \frac{V_0}{R(1 - aRC)} \left[e^{-at} - e^{-\frac{t}{RC}} \right] \text{ A} \quad \left(\text{for } a \neq \frac{1}{RC} \right)$$

3.2 2082 Bhadra — Question 3 [6 Marks]

Problem Statement (2082 Bhadra, Q3)

Derive the differential equation of series RLC circuit when excited by step voltage and discuss the different cases of circuit responses.

Detailed Derivation & Damping Analysis

1. Circuit Differential Equation: For a series RLC circuit excited by step voltage $V_0 u(t)$:

$$Ri(t) + L \frac{di}{dt} + \frac{1}{C} \int i(t) dt = V_0$$

Differentiating with respect to t and dividing by L :

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i(t) = 0 \iff s^2 + 2\alpha s + \omega_0^2 = 0 \quad (33)$$

where $\alpha = \frac{R}{2L}$ (damping attenuation) and $\omega_0 = \frac{1}{\sqrt{LC}}$ (undamped natural frequency). The roots are:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \quad (34)$$

2. The Three Damping Response Cases:

1. **Overdamped Case** ($\alpha > \omega_0 \iff R > 2\sqrt{L/C}$): Roots s_1, s_2 are real, distinct, and negative. Current rises smoothly and decays exponentially without any oscillation:

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (35)$$

2. **Critically Damped Case** ($\alpha = \omega_0 \iff R = 2\sqrt{L/C}$): Roots are real and equal ($s_1 = s_2 = -\alpha$). Circuit achieves steady state in the minimum possible transition time without overshoot:

$$i(t) = (A_1 + A_2 t) e^{-\alpha t} \quad (36)$$

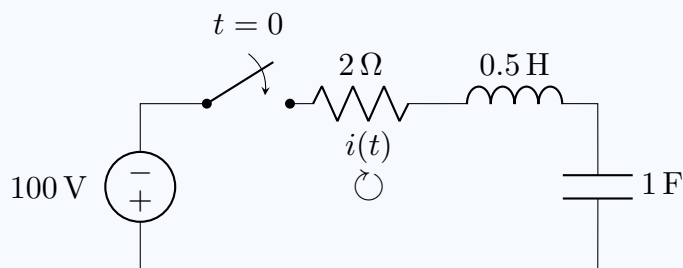
3. **Underdamped Case** ($\alpha < \omega_0 \iff R < 2\sqrt{L/C}$): Roots are complex conjugate pairs $s_{1,2} = -\alpha \pm j\omega_d$, where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ is the damped natural frequency. Response exhibits exponentially decaying sinusoidal ringing:

$$i(t) = e^{-\alpha t} [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)] \quad (37)$$

3.3 2082 Baishakh — Question 3 [6 Marks]

Problem Statement (2082 Baishakh, Q3)

A DC source of 100 V is suddenly applied at time $t = 0$ to a series RLC circuit comprising $R = 2\ \Omega$, $L = 0.5\ \text{H}$ and $C = 1\ \text{F}$. Obtain the expression for current in the circuit by using classical method:



Detailed Numerical Solution

1. Characteristic Equation:

$$0.5 \frac{d^2 i}{dt^2} + 2 \frac{di}{dt} + \frac{1}{1} i = 0 \implies \frac{d^2 i}{dt^2} + 4 \frac{di}{dt} + 2i = 0$$

Characteristic roots:

$$s^2 + 4s + 2 = 0 \implies s_{1,2} = \frac{-4 \pm \sqrt{16 - 8}}{2} = -2 \pm \sqrt{2}$$

$s_1 = -2 + \sqrt{2} \approx -0.5858\ \text{s}^{-1}$ and $s_2 = -2 - \sqrt{2} \approx -3.4142\ \text{s}^{-1}$ (Overdamped).

2. General Solution:

$$i(t) = A_1 e^{(-2+\sqrt{2})t} + A_2 e^{(-2-\sqrt{2})t}$$

3. Boundary Conditions:

- At $t = 0^+$: $i(0^+) = 0 \implies A_1 + A_2 = 0 \implies A_2 = -A_1$.

- From KVL at $t = 0^+$:

$$100 = Ri(0^+) + L \left. \frac{di}{dt} \right|_{0^+} + v_C(0^+) = 0 + 0.5 \left. \frac{di}{dt} \right|_{0^+} + 0 \implies \left. \frac{di}{dt} \right|_{0^+} = 200\ \text{A/s}$$

- Differentiating $i(t)$ at $t = 0^+$:

$$s_1 A_1 + s_2 A_2 = 200 \implies A_1(s_1 - s_2) = 200 \implies A_1(2\sqrt{2}) = 200 \implies A_1 = 50\sqrt{2} \approx 70.71$$

$$A_2 = -50\sqrt{2} \approx -70.71.$$

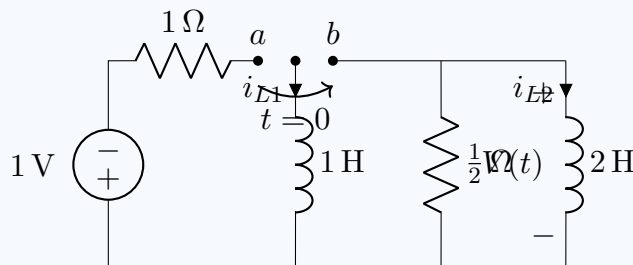
Verified Final Result

$$i(t) = 50\sqrt{2} \left[e^{(-2+\sqrt{2})t} - e^{(-2-\sqrt{2})t} \right] \text{ A} = 70.71 (e^{-0.586t} - e^{-3.414t}) \text{ A} \quad (t \geq 0)$$

3.4 2081 Ashwin — Question 3 [6 Marks]

Problem Statement (2081 Ashwin, Q3)

The circuit shown below is in steady state with switch at position 'a'. The switch is moved to position 'b' at $t = 0$. Find the expression for $V(t)$ and current through 2H inductor $i_L(t)$ for $t > 0$ using classical method:



Detailed Step-by-Step Solution

1. Initial Conditions at $t = 0^-$ (Switch at 'a'): In DC steady state, the 1 H inductor acts as a short circuit:

$$i_{L1}(0^-) = \frac{1\text{ V}}{1\ \Omega} = 1\text{ A} \quad (\text{downward}), \quad i_{L2}(0^-) = 0\text{ A}$$

By continuity, at $t = 0^+$: $i_{L1}(0^+) = 1\text{ A}$, $i_{L2}(0^+) = 0\text{ A}$.

2. Parallel Network Analysis for $t > 0$ (Switch at 'b'): When thrown to 'b', the 1 V source and 1 Ω resistor are isolated. The remaining network consists of $L_1 = 1\text{ H}$, $R = 0.5\ \Omega$, and $L_2 = 2\text{ H}$ in parallel across node voltage $V(t)$.

Applying KCL at the top node:

$$\frac{V(t)}{R} + \left[i_{L1}(0^+) + \frac{1}{L_1} \int_0^t V(\tau) d\tau \right] + \left[i_{L2}(0^+) + \frac{1}{L_2} \int_0^t V(\tau) d\tau \right] = 0 \quad (38)$$

Substitute $R = 0.5\ \Omega$, $L_1 = 1\text{ H}$, $L_2 = 2\text{ H}$, $i_{L1}(0^+) = 1\text{ A}$, $i_{L2}(0^+) = 0$:

$$2V(t) + 1 + \left(1 + \frac{1}{2} \right) \int_0^t V(\tau) d\tau = 0 \implies 2V(t) + \frac{3}{2} \int_0^t V(\tau) d\tau = -1$$

At $t = 0^+$, the integral is zero $\implies 2V(0^+) = -1 \implies V(0^+) = -0.5\text{ V}$.

Differentiating with respect to t :

$$2 \frac{dV}{dt} + \frac{3}{2} V(t) = 0 \implies \frac{dV}{dt} + \frac{3}{4} V(t) = 0 \quad (39)$$

Solving this first-order linear ODE:

$$V(t) = V(0^+) e^{-\frac{3}{4}t} = -0.5 e^{-0.75t} \text{ V} \quad (t > 0) \quad (40)$$

3. Current through 2H Inductor $i_{L2}(t)$:

$$\begin{aligned} i_{L2}(t) &= i_{L2}(0^+) + \frac{1}{L_2} \int_0^t V(\tau) d\tau = 0 + \frac{1}{2} \int_0^t (-0.5 e^{-0.75\tau}) d\tau \\ &= -0.25 \left[\frac{e^{-0.75\tau}}{-0.75} \right]_0^t = \frac{0.25}{0.75} (e^{-0.75t} - 1) = \frac{1}{3} (e^{-0.75t} - 1) \text{ A} \\ &= -0.333 (1 - e^{-0.75t}) \text{ A} \quad (t > 0) \end{aligned} \quad (41)$$

Verified Final Result**Node Voltage:**

$$V(t) = -0.5e^{-0.75t} \text{ V} \quad (t > 0)$$

Inductor Current (2 H):

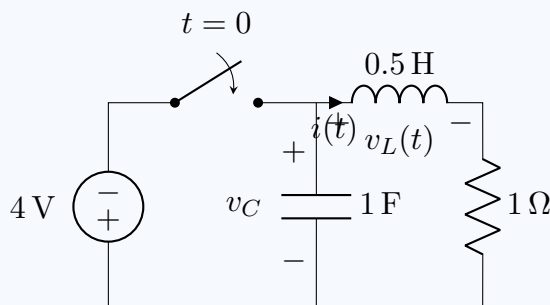
$$i_{L2}(t) = \frac{1}{3} (e^{-0.75t} - 1) \text{ A} = -0.333 (1 - e^{-0.75t}) \text{ A} \quad (t > 0)$$

4 Question 4: Laplace Transform Applications to Circuit Analysis

4.1 2083 Baishakh — Question 4 [6 Marks]

Problem Statement (2083 Baishakh, Q4)

In the given circuit, switch is opened at $t = 0$. Find the current and voltage across inductor using Laplace transform method:



Detailed Step-by-Step Solution

1. Initial Conditions ($t = 0^-$): Before $t = 0$, the switch is closed. In DC steady state:

- Inductor is a short circuit $\Rightarrow i_L(0^-) = \frac{4V}{1\Omega} = 4\text{ A}$ (downward).
- Capacitor is an open circuit $\Rightarrow v_C(0^-) = 4\text{ V}$ (top positive).

By continuity: $i_L(0^+) = 4\text{ A}$, $v_C(0^+) = 4\text{ V}$.

2. s -Domain Formulation for $t \geq 0$: When the switch opens at $t = 0$, the 4 V supply is disconnected. The remaining network is a single closed loop containing:

- Capacitor $C = 1\text{ F}$ with initial voltage generator $\frac{V_0}{s} = \frac{4}{s}$ in series with $\frac{1}{sC} = \frac{1}{s}$.
- Inductor $L = 0.5\text{ H}$ with initial current generator $Li_L(0^+) = 0.5 \times 4 = 2\text{ V}$ opposing current in series with $sL = 0.5s$.
- Resistor $R = 1\Omega$.

Applying KVL in the s -domain around the loop (clockwise current $I(s)$):

$$\begin{aligned} \frac{4}{s} - \frac{1}{s}I(s) &= (0.5s + 1)I(s) - 2 \\ \frac{4 + 2s}{s} &= \left(0.5s + 1 + \frac{1}{s}\right)I(s) = \frac{0.5s^2 + s + 1}{s}I(s) \\ I(s) &= \frac{2s + 4}{0.5s^2 + s + 1} = \frac{4s + 8}{s^2 + 2s + 2} \end{aligned} \quad (42)$$

3. Inverse Laplace Transform for $i(t)$: Completing the square in the denominator: $s^2 + 2s + 2 = (s + 1)^2 + 1^2$:

$$\begin{aligned} I(s) &= \frac{4(s + 1) + 4}{(s + 1)^2 + 1^2} = 4 \frac{s + 1}{(s + 1)^2 + 1^2} + 4 \frac{1}{(s + 1)^2 + 1^2} \\ i(t) &= 4e^{-t} \cos(t) + 4e^{-t} \sin(t) = 4e^{-t}(\cos t + \sin t) \text{ A} \quad (t \geq 0) \end{aligned} \quad (43)$$

4. Inductor Voltage $v_L(t)$: In the s -domain:

$$\begin{aligned} V_L(s) &= sLI(s) - Li_L(0^+) = 0.5s \left(\frac{4s+8}{s^2+2s+2} \right) - 2 \\ &= \frac{2s^2+4s-2(s^2+2s+2)}{s^2+2s+2} = \frac{-4}{s^2+2s+2} = -4 \frac{1}{(s+1)^2+1^2} \end{aligned} \quad (44)$$

Taking the Inverse Laplace Transform:

$$v_L(t) = -4e^{-t} \sin(t) \text{ V} \quad (t \geq 0) \quad (45)$$

Verification: At $t = 0^+$, $v_L(0^+) = 0 \text{ V} = v_C(0^+) - Ri(0^+) = 4 - (1)(4) = 0 \text{ V}$.

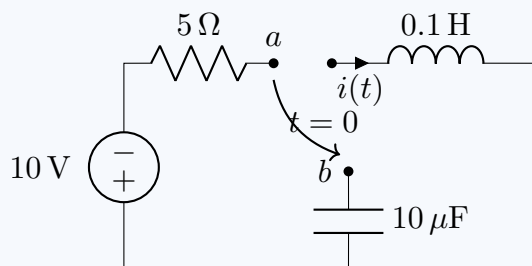
Verified Final Result

$$i(t) = 4e^{-t}(\cos t + \sin t) \text{ A}, \quad v_L(t) = -4e^{-t} \sin(t) \text{ V} \quad (t \geq 0)$$

4.2 2082 Bhadra — Question 4 [6 Marks]

Problem Statement (2082 Bhadra, Q4)

In the circuit shown switch is changed from position a to position b at $t = 0$. Determine the current $i(t)$ using Laplace transform method:



Step-by-Step Solution

1. Initial Conditions ($t = 0^-$): For $t < 0$, switch is at 'a', connecting 10 V source across 5Ω resistor and 0.1 H inductor:

$$i_L(0^-) = \frac{10\text{ V}}{5\Omega} = 2\text{ A}, \quad v_C(0^-) = 0\text{ V} \quad (\text{terminal } b \text{ unconnected})$$

2. s -Domain Circuit ($t > 0$): At $t = 0$, moving to 'b' creates a series LC loop of $L = 0.1\text{ H}$ ($i_L(0^+) = 2\text{ A}$) and $C = 10\mu\text{F} = 10^{-5}\text{ F}$ ($v_C(0^+) = 0$).

$$\left(sL + \frac{1}{sC}\right) I(s) = Li_L(0^+) \implies \left(0.1s + \frac{10^5}{s}\right) I(s) = 0.1 \times 2 = 0.2 \quad (46)$$

Multiplying by s :

$$(0.1s^2 + 10^5)I(s) = 0.2s \implies I(s) = \frac{0.2s}{0.1(s^2 + 10^6)} = \frac{2s}{s^2 + (1000)^2} \quad (47)$$

3. Inverse Laplace Transform:

$$i(t) = \mathcal{L}^{-1} \left\{ \frac{2s}{s^2 + 1000^2} \right\} = 2 \cos(1000t) \text{ A} \quad (t \geq 0) \quad (48)$$

Verified Final Result

$$i(t) = 2 \cos(1000t) \text{ A} \quad (t \geq 0)$$

4.3 2082 Baishakh — Question 4 [6 Marks]

Problem Statement (2082 Baishakh, Q4)

An exponential voltage $V(t) = 20e^{-4t}$ is suddenly applied at time $t = 0$ to series RLC circuit comprising $R = 2\ \Omega$, $L = 0.5\text{ H}$ and $C = 1\text{ F}$. Obtain the expression for the current $i(t)$ in the circuit using Laplace method.

Step-by-Step Solution

1. **s-Domain Impedance & Excitation:** $V(s) = \frac{20}{s+4}$, $Z(s) = R + sL + \frac{1}{sC} = 2 + 0.5s + \frac{1}{s} = \frac{s^2 + 4s + 2}{2s}$.

$$I(s) = \frac{V(s)}{Z(s)} = \frac{\frac{20}{s+4}}{\frac{s^2 + 4s + 2}{2s}} = \frac{40s}{(s+4)(s^2 + 4s + 2)} \quad (49)$$

2. **Partial Fraction Decomposition:**

$$\frac{40s}{(s+4)(s^2 + 4s + 2)} = \frac{A}{s+4} + \frac{Bs + C}{s^2 + 4s + 2}$$

- $A = \left. \frac{40s}{s^2 + 4s + 2} \right|_{s=-4} = \frac{-160}{16 - 16 + 2} = -80$.
- $40s = -80(s^2 + 4s + 2) + (Bs + C)(s + 4) \implies B = 80, \quad C = 40$.

3. **Inverse Laplace Transform:**

$$I(s) = \frac{-80}{s+4} + \frac{80s + 40}{(s+2)^2 - 2} = \frac{-80}{s+4} + 80 \frac{s+2}{(s+2)^2 - (\sqrt{2})^2} - \frac{120}{\sqrt{2}} \frac{\sqrt{2}}{(s+2)^2 - (\sqrt{2})^2}$$

Taking $\mathcal{L}^{-1}$:

$$i(t) = -80e^{-4t} + 80e^{-2t} \cosh(\sqrt{2}t) - 60\sqrt{2}e^{-2t} \sinh(\sqrt{2}t) \text{ A} \quad (t \geq 0) \quad (50)$$

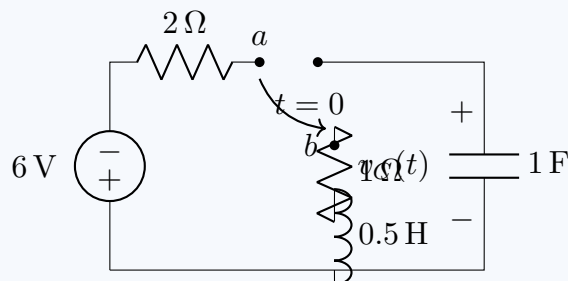
Verified Final Result

$$i(t) = -80e^{-4t} + 80e^{-2t} \cosh(\sqrt{2}t) - 84.85e^{-2t} \sinh(\sqrt{2}t) \text{ A} \quad (t \geq 0)$$

4.4 2081 Ashwin — Question 4 [6 Marks]

Problem Statement (2081 Ashwin, Q4)

The switch in the figure below has been in position 'a' for a long time. Then, it is moved to 'b' at $t = 0$. Obtain the expression for voltage across capacitor for $t > 0$ using Laplace transform method:



Step-by-Step Solution

1. **Initial Conditions** ($t = 0^-$): Switch at 'a': $v_C(0^-) = 6\text{ V}$, $i_L(0^-) = 0\text{ A}$.

2. **s-Domain Analysis** ($t > 0$): Series RLC discharging loop ($R = 1\ \Omega$, $L = 0.5\text{ H}$, $C = 1\text{ F}$, $V_0 = 6\text{ V}$):

$$\left(0.5s + 1 + \frac{1}{s}\right) I(s) = \frac{6}{s} \Rightarrow I(s) = \frac{12}{s^2 + 2s + 2}$$

Capacitor voltage in s -domain:

$$\begin{aligned} V_C(s) &= \frac{v_C(0^+)}{s} - \frac{1}{sC} I(s) = \frac{6}{s} - \frac{12}{s(s^2 + 2s + 2)} = \frac{6s + 12}{s^2 + 2s + 2} \\ &= \frac{6(s+1) + 6}{(s+1)^2 + 1^2} = 6 \frac{s+1}{(s+1)^2 + 1^2} + 6 \frac{1}{(s+1)^2 + 1^2} \end{aligned}$$

Taking $\mathcal{L}^{-1}$:

$$v_C(t) = 6e^{-t}(\cos t + \sin t)\text{ V} \quad (t > 0) \quad (51)$$

Verified Final Result

$$v_C(t) = 6e^{-t}(\cos t + \sin t)\text{ V} \quad (t > 0)$$

5 Questions 5 & 6: Frequency Response & Asymptotic Bode Plots

5.1 2083 Baishakh — Question 5 [7 Marks]

Problem Statement (2083 Baishakh, Q5)

What is network function? Plot Asymptotic Bode graph of given transfer function:

$$H(s) = \frac{10(s^2 + 8s + 15)}{s^2(s + 3)(0.05s + 1)}$$

Solution & Complete Plot

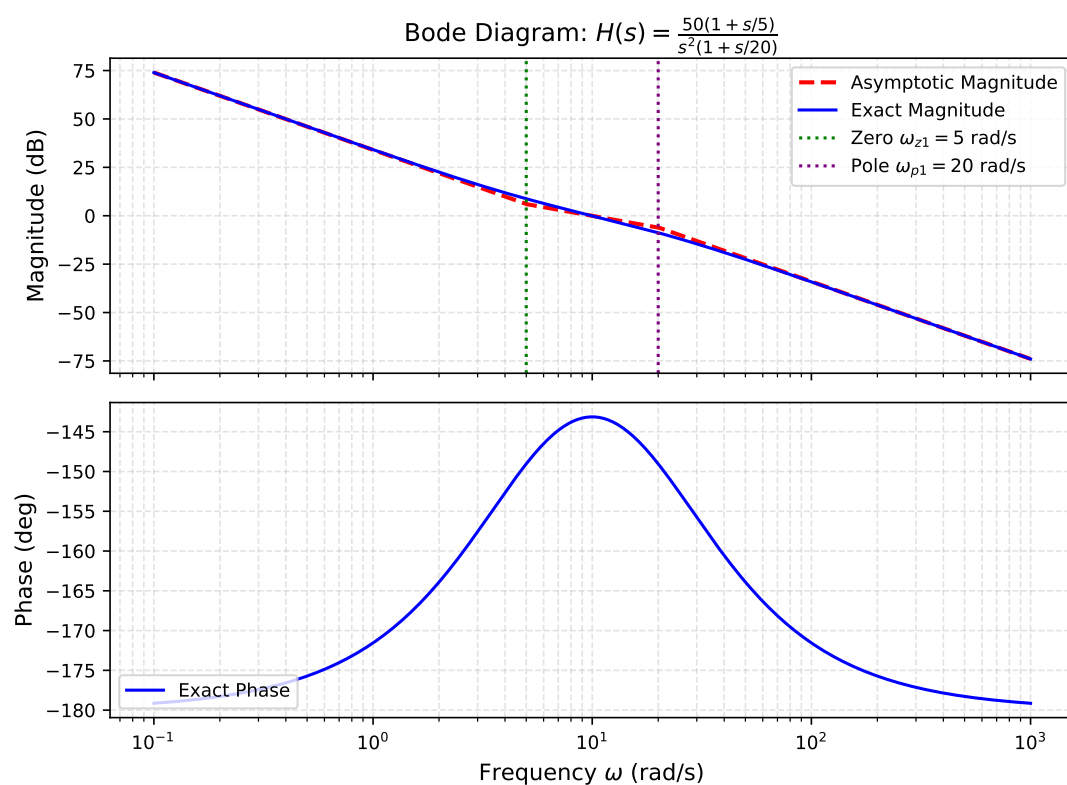
1. Network Function Definition: A network function $H(s) = \frac{Y(s)}{X(s)}$ is the ratio of the Laplace transform of the output response $Y(s)$ to the Laplace transform of the input excitation $X(s)$, with all initial conditions set to zero.

2. Factorization into Standard Bode Form: Factor numerator: $s^2 + 8s + 15 = (s + 3)(s + 5)$. Canceling the common pole-zero term $(s + 3)$:

$$H(s) = \frac{10(s + 5)}{s^2(0.05s + 1)} = \frac{10 \times 5 \left(1 + \frac{s}{5}\right)}{s^2 \left(1 + \frac{s}{20}\right)} = \frac{50 \left(1 + \frac{s}{5}\right)}{s^2 \left(1 + \frac{s}{20}\right)}$$

- **DC/System Gain:** $K = 50 \implies 20 \log_{10}(50) \approx 33.98 \text{ dB}$.
- **Type-2 System (Double Pole at Origin):** Initial low-frequency slope of -40 dB/dec .
- **Corner Frequencies:** Simple zero at $\omega_{z1} = 5 \text{ rad/s}$ ($+20 \text{ dB/dec}$ slope change); Simple pole at $\omega_{p1} = 20 \text{ rad/s}$ (-20 dB/dec slope change).

Frequency Range (rad/s)	Active Factors	Slope Change	Resultant Slope
$\omega < 5$	$50/s^2$	—	-40 dB/decade
$5 \leq \omega < 20$	Zero at $\omega = 5$	$+20 \text{ dB/dec}$	-20 dB/decade
$\omega \geq 20$	Pole at $\omega = 20$	-20 dB/dec	-40 dB/decade



5.2 2082 Bhadra — Question 6 [1 + 7 = 8 Marks]

Problem Statement (2082 Bhadra, Q6)

Define poles and zeros of network function. Construct the asymptotic magnitude bode plot and phase plot for given transfer function:

$$T(s) = \frac{100(s+5)}{s^2(s^2+21s+20)}$$

Solution & Complete Plot

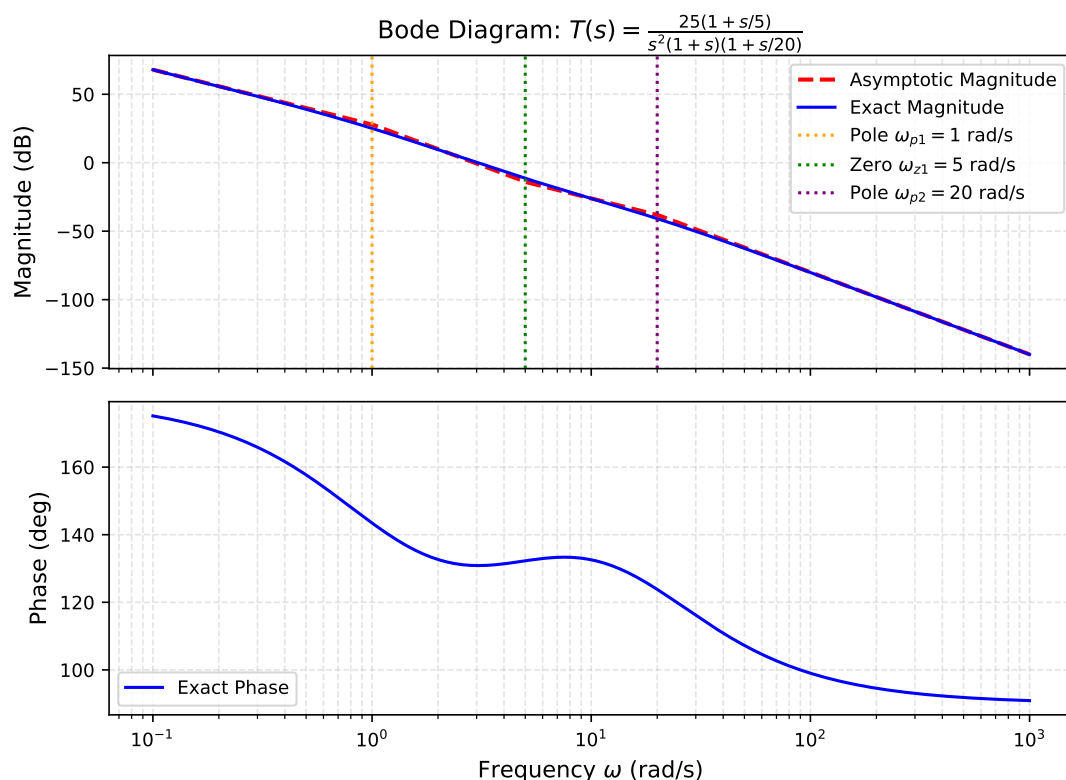
1. Definitions: **Poles** are the complex frequencies $s = p_k$ where $T(s) \rightarrow \infty$ (roots of denominator). **Zeros** are the complex frequencies $s = z_k$ where $T(s) \rightarrow 0$ (roots of numerator).

2. Factorization into Standard Bode Form: Denominator quadratic: $s^2+21s+20 = (s+1)(s+20)$.

$$T(s) = \frac{100 \times 5 \left(1 + \frac{s}{5}\right)}{s^2 \times 1(1+s) \times 20 \left(1 + \frac{s}{20}\right)} = \frac{25 \left(1 + \frac{s}{5}\right)}{s^2(1+s) \left(1 + \frac{s}{20}\right)}$$

- $K = 25 \Rightarrow 20 \log_{10}(25) \approx 27.96 \text{ dB}$.
- Initial slope: -40 dB/dec (double pole at origin).
- Corner frequencies in ascending order: $\omega_{p1} = 1 \text{ rad/s} < \omega_{z1} = 5 \text{ rad/s} < \omega_{p2} = 20 \text{ rad/s}$.

Frequency Range (rad/s)	Factor Introduced	Slope Change	Resultant Slope
$\omega < 1$	$25/s^2$	—	-40 dB/decade
$1 \leq \omega < 5$	Pole at $\omega = 1$	-20 dB/dec	-60 dB/decade
$5 \leq \omega < 20$	Zero at $\omega = 5$	$+20 \text{ dB/dec}$	-40 dB/decade
$\omega \geq 20$	Pole at $\omega = 20$	-20 dB/dec	-60 dB/decade



5.3 2082 Baishakh — Question 5 [8 Marks]

Problem Statement (2082 Baishakh, Q5)

Sketch the asymptotic bode plot for the transfer function given by:

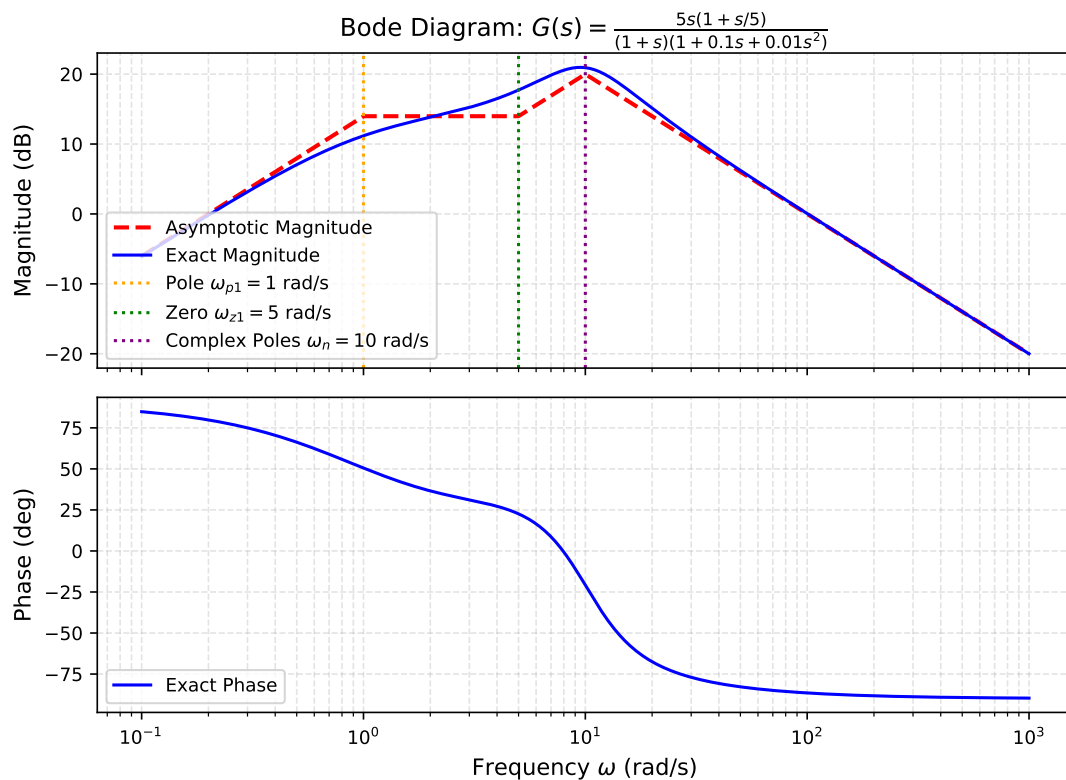
$$G(s) = \frac{100s(s+5)}{(s+1)(s^2+10s+100)}$$

Solution & Complete Plot

Standard Bode Form:

$$G(s) = \frac{100 \cdot s \cdot 5 \left(1 + \frac{s}{5}\right)}{1(1+s) \cdot 100 \left(1 + \frac{s}{10} + \frac{s^2}{100}\right)} = \frac{5s \left(1 + \frac{s}{5}\right)}{(1+s)(1+0.1s+0.01s^2)}$$

- Zero at origin gives initial low-frequency slope of +20 dB/dec with $20 \log_{10}(5\omega)$.
- Simple pole at $\omega_{p1} = 1$ rad/s (-20 dB/dec $\Rightarrow$ slope 0 dB/dec).
- Simple zero at $\omega_{z1} = 5$ rad/s ($+20$ dB/dec $\Rightarrow$ slope +20 dB/dec).
- Complex conjugate poles at $\omega_n = \sqrt{100} = 10$ rad/s ($\zeta = 0.5 < 1$) introduce -40 dB/dec slope change $\Rightarrow$ net slope -20 dB/dec.



5.4 2081 Ashwin — Question 5 [7 Marks]

Problem Statement (2081 Ashwin, Q5)

Plot asymptotic Bode graph of given transfer function:

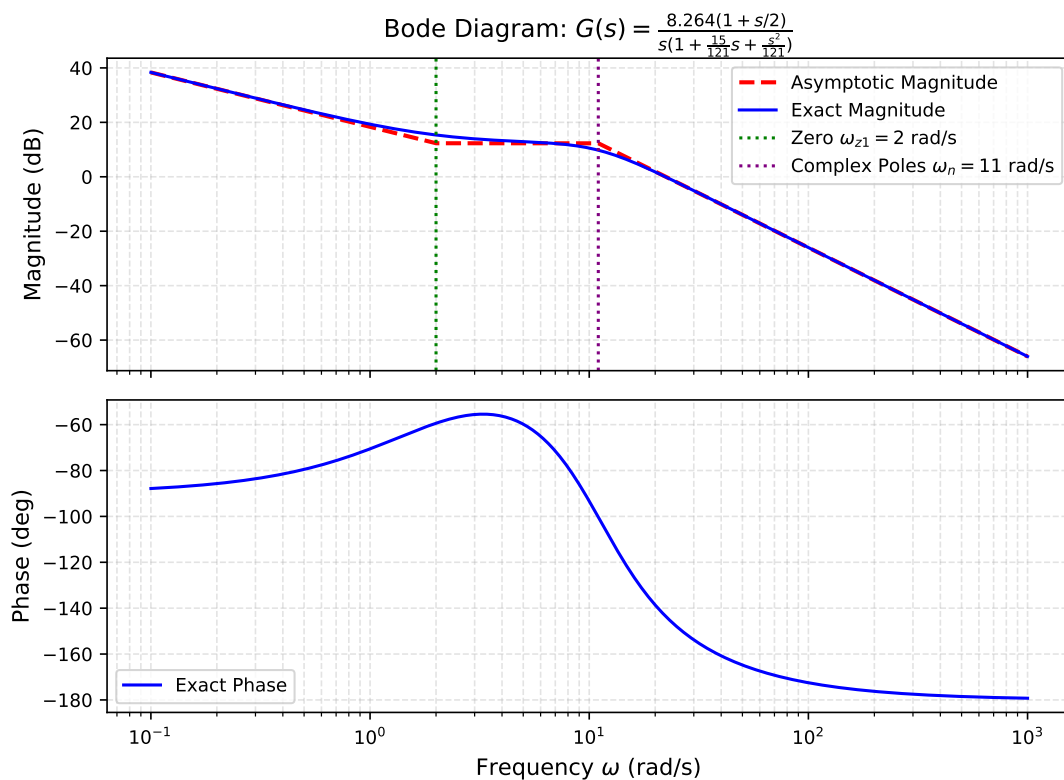
$$G(s) = \frac{500(s+2)}{s(s^2+15s+121)}$$

Solution & Complete Plot

Standard Bode Form:

$$G(s) = \frac{500 \times 2 \left(1 + \frac{s}{2}\right)}{s \times 121 \left(1 + \frac{15}{121}s + \frac{s^2}{121}\right)} = \frac{8.264 \left(1 + \frac{s}{2}\right)}{s \left(1 + \frac{15}{121}s + \frac{s^2}{121}\right)}$$

- $K = 8.264 \Rightarrow 20 \log_{10}(8.264) \approx 18.34 \text{ dB}$.
- Initial slope: -20 dB/dec due to pole at origin.
- Simple zero at $\omega_{z1} = 2 \text{ rad/s}$ ($+20 \text{ dB/dec} \Rightarrow \text{slope } 0 \text{ dB/dec}$).
- Complex conjugate pole pair at $\omega_n = \sqrt{121} = 11 \text{ rad/s}$ ($\zeta = \frac{15}{22} = 0.682 < 1$) introduces $-40 \text{ dB/dec} \Rightarrow \text{net slope } -40 \text{ dB/dec}$.

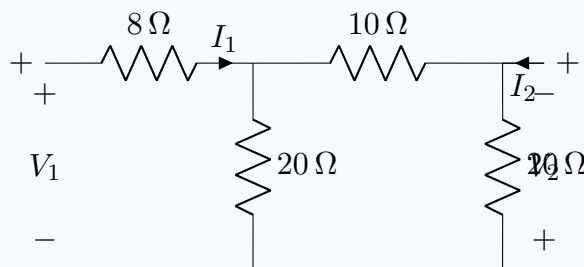


6 Question 6: Two-Port Network Parameters & Interconnections

6.1 2083 Baishakh — Question 6 [6 Marks]

Problem Statement (2083 Baishakh, Q6)

Find Z parameter for given two port network and state whether network is symmetrical or reciprocal:



Step-by-Step Derivation

1. Open-Circuit at Port 2 ($I_2 = 0$): With Port 2 open, the second 20Ω resistor is in series with the 10Ω resistor, forming a 30Ω branch in parallel with the middle 20Ω resistor:

$$R_{\text{parallel}} = 20 \parallel (10 + 20) = 20 \parallel 30 = \frac{600}{50} = 12\Omega$$

- $Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} = 8 + 12 = 20\Omega.$

- Voltage across middle node: $V_m = 12I_1$. By voltage division across $(10 + 20)\Omega$:

$$V_2 = V_m \left(\frac{20}{10 + 20} \right) = (12I_1) \left(\frac{20}{30} \right) = 8I_1 \implies Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0} = 8\Omega$$

2. Open-Circuit at Port 1 ($I_1 = 0$): With Port 1 open, no current flows through the 8Ω resistor.

- $Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = 20 \parallel (10 + 20) = 12\Omega.$

- By current division, current entering the $(10 + 20)\Omega$ branch is $I_{\text{branch}} = I_2 \frac{20}{20+30} = 0.4I_2$.

$$V_1 = I_{\text{branch}} \times 20\Omega = 0.4I_2 \times 20 = 8I_2 \implies Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0} = 8\Omega$$

3. Symmetry & Reciprocity:

- **Reciprocity:** $Z_{12} = Z_{21} = 8\Omega \implies$ **Reciprocal.**
- **Symmetry:** $Z_{11} = 20\Omega \neq 12\Omega = Z_{22} \implies$ **Not Symmetrical.**

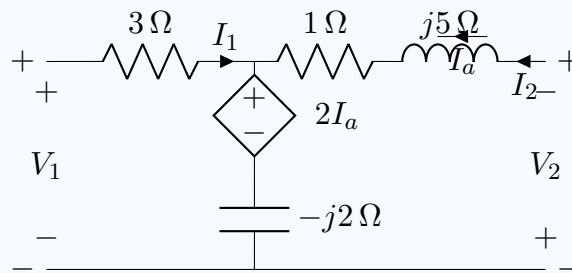
Verified Final Result

$$[\mathbf{Z}] = \begin{bmatrix} 20 & 8 \\ 8 & 12 \end{bmatrix} \Omega \quad \text{Network is } \mathbf{Reciprocal} \text{ but } \mathbf{Not Symmetrical}.$$

6.2 2082 Bhadra — Question 5 [5 Marks]

Problem Statement (2082 Bhadra, Q5)

Define two port network. For the given network, find Z parameters:



Step-by-Step Solution

1. Controlling Current Relation: The current I_a is labeled flowing leftward into Port 2, which matches the standard terminal current I_2 :

$$I_a = I_2 \quad (52)$$

2. KVL Equations: Current flowing downward through the middle shunt branch is $(I_1 + I_2)$. The voltage across this branch is:

$$V_{\text{shunt}} = 2I_a + (-j2)(I_1 + I_2) = 2I_2 - j2I_1 - j2I_2 = -j2I_1 + (2 - j2)I_2$$

• **Port 1 KVL:**

$$V_1 = 3I_1 + V_{\text{shunt}} = (3 - j2)I_1 + (2 - j2)I_2 \implies Z_{11} = 3 - j2 \Omega, \quad Z_{12} = 2 - j2 \Omega$$

• **Port 2 KVL:**

$$\begin{aligned} V_2 &= (1 + j5)I_2 + V_{\text{shunt}} = -j2I_1 + (1 + j5 + 2 - j2)I_2 = -j2I_1 + (3 + j3)I_2 \\ \implies Z_{21} &= -j2 \Omega, \quad Z_{22} = 3 + j3 \Omega \end{aligned}$$

Verified Final Result

$$[\mathbf{Z}] = \begin{bmatrix} 3 - j2 & 2 - j2 \\ -j2 & 3 + j3 \end{bmatrix} \Omega \quad (\text{Non-reciprocal due to dependent source})$$

6.3 2082 Baishakh — Question 6 [3 + 3 = 6 Marks]

Problem Statement (2082 Baishakh, Q6)

Find the expression of hybrid parameters in terms of Z parameters. Derive the equivalent T parameters of two cascade connected two port networks.

Part A: Hybrid (h) Parameters in terms of Z Parameters

Governing equations:

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \text{--- (1),} \quad V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \text{--- (2)} \quad (53)$$

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \text{--- (3),} \quad I_2 = h_{21}I_1 + h_{22}V_2 \quad \text{--- (4)} \quad (54)$$

From (2), solve for I_2 :

$$I_2 = -\frac{Z_{21}}{Z_{22}}I_1 + \frac{1}{Z_{22}}V_2 \implies h_{21} = -\frac{Z_{21}}{Z_{22}}, \quad h_{22} = \frac{1}{Z_{22}} \quad (55)$$

Substitute I_2 into (1):

$$\begin{aligned} V_1 &= Z_{11}I_1 + Z_{12}\left(-\frac{Z_{21}}{Z_{22}}I_1 + \frac{1}{Z_{22}}V_2\right) = \left(\frac{Z_{11}Z_{22} - Z_{12}Z_{21}}{Z_{22}}\right)I_1 + \left(\frac{Z_{12}}{Z_{22}}\right)V_2 \\ \implies h_{11} &= \frac{\Delta_Z}{Z_{22}}, \quad h_{12} = \frac{Z_{12}}{Z_{22}} \quad (\text{where } \Delta_Z = Z_{11}Z_{22} - Z_{12}Z_{21}) \end{aligned} \quad (56)$$

Part B: Transmission ($T/ABCD$) Parameters of Cascaded Networks

For two networks N_A and N_B in cascade:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \mathbf{T}_A \begin{bmatrix} V_2' \\ -I_2' \end{bmatrix}, \quad \begin{bmatrix} V_1'' \\ I_1'' \end{bmatrix} = \mathbf{T}_B \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

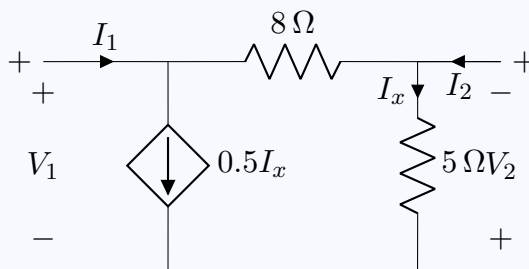
At the cascade interface: $V_2' = V_1''$ and $-I_2' = I_1''$. Therefore:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \mathbf{T}_A \mathbf{T}_B \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \implies \mathbf{T}_{eq} = \mathbf{T}_A \times \mathbf{T}_B = \begin{bmatrix} A_A A_B + B_A C_B & A_A B_B + B_A D_B \\ C_A A_B + D_A C_B & C_A B_B + D_A D_B \end{bmatrix} \quad (57)$$

6.4 2081 Ashwin — Question 6 [2 + 4 = 6 Marks]

Problem Statement (2081 Ashwin, Q6)

Find the expression of Z parameters in terms of ABCD parameters. For given two port networks, find ABCD parameters:



Part A: Z-Parameters in terms of ABCD Parameters

$$V_1 = AV_2 - BI_2 \quad \text{--- (1)}, \quad I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

From (2): $V_2 = \frac{1}{C}I_1 + \frac{D}{C}I_2 \Rightarrow Z_{21} = \frac{1}{C}, \quad Z_{22} = \frac{D}{C}$. Substituting V_2 into (1):

$$V_1 = A\left(\frac{1}{C}I_1 + \frac{D}{C}I_2\right) - BI_2 = \frac{A}{C}I_1 + \left(\frac{AD - BC}{C}\right)I_2 \Rightarrow Z_{11} = \frac{A}{C}, \quad Z_{12} = \frac{\Delta_T}{C}$$

Part B: Calculation of ABCD Parameters for the Active Network

1. The 5Ω resistor is connected directly across Port 2:

$$V_2 = 5I_x \Rightarrow I_x = 0.2V_2 \quad (58)$$

2. At the Port 2 top node, current entering from left through 8Ω resistor is I_{series} :

$$I_{\text{series}} + I_2 = I_x \Rightarrow I_{\text{series}} = I_x - I_2 = 0.2V_2 - I_2 \quad (59)$$

3. Port 1 Voltage V_1 :

$$V_1 = V_2 + 8I_{\text{series}} = V_2 + 8(0.2V_2 - I_2) = 2.6V_2 - 8I_2 \Rightarrow A = 2.6, \quad B = 8\Omega \quad (60)$$

4. Port 1 Current I_1 :

$$I_1 = 0.5I_x + I_{\text{series}} = 0.5(0.2V_2) + (0.2V_2 - I_2) = 0.3V_2 - I_2 \Rightarrow C = 0.3\text{ S}, \quad D = 1 \quad (61)$$

Verified Final Result

$$\mathbf{T} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 2.6 & 8\Omega \\ 0.3\text{ S} & 1 \end{bmatrix}, \quad \Delta_T = AD - BC = (2.6)(1) - (8)(0.3) = 0.2 \neq 1$$

7 Question 7: Single-Phase Transformers (Theory & Testing)

7.1 2083 Baishakh Q7 & 2081 Ashwin Q7 [5–6 Marks]

Problem Statement (2083 Baishakh Q7 / 2081 Ashwin Q7)

List out features of ideal transformer. Explain operation of transformer under loaded condition and show/prove that the main magnetic flux in the transformer core is constant and independent of load current.

1. Features of an Ideal Transformer

- **Zero Winding Resistance:** $R_1 = 0, R_2 = 0 \implies$ zero ohmic copper loss ($I^2R = 0$).
- **Infinite Core Permeability** ($\mu_r \rightarrow \infty$): Zero reluctance $\implies$ vanishingly small magnetizing current ($I_m \rightarrow 0$) needed to establish core flux.
- **Zero Leakage Flux:** 100% magnetic coupling ($k = 1$); all flux is mutual core flux Φ_m .
- **Zero Core Losses:** No hysteresis or eddy current losses in the core.
- **100% Efficiency:** $P_{in} = P_{out}$ under all loading conditions.

2. Proof that Main Core Flux Φ_m is Constant Under All Loads

1. **No-Load Equilibrium:** When energized by primary sinusoidal voltage V_1 at frequency f , primary winding draws no-load current I_0 establishing peak flux Φ_m . By Faraday's Law, the induced back EMF is:

$$E_1 = 4.44fN_1\Phi_m \implies \Phi_m = \frac{E_1}{4.44fN_1} \quad (62)$$

Since primary winding impedance drops are negligible ($V_1 \approx E_1$):

$$\Phi_m \approx \frac{V_1}{4.44fN_1} = \text{Constant} \quad (63)$$

2. **Action Under Load:** When secondary is connected to a load, current I_2 flows, creating a demagnetizing secondary MMF N_2I_2 producing opposing flux Φ_2 .
3. **Self-Balancing Primary Reaction:** The net flux momentarily dips, reducing primary back EMF E_1 . The resulting voltage difference ($V_1 - E_1$) instantly draws an additional primary current I'_2 (load component) such that:

$$N_1I'_2 = N_2I_2 \implies \Phi'_2 = \Phi_2 \quad (64)$$

The primary load flux Φ'_2 completely neutralizes the secondary demagnetizing flux Φ_2 .

4. **Conclusion:** The net core MMF remains N_1I_0 , keeping core flux Φ_m strictly constant.

7.2 2082 Bhadra — Question 7 [5 Marks]

Problem Statement (2082 Bhadra, Q7)

State the role of commutator and carbon brushes in DC machine. Explain the characteristics of DC generators.

1. Role of Commutator and Carbon Brushes

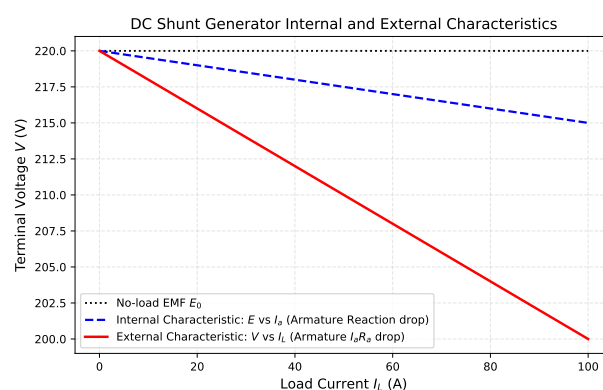
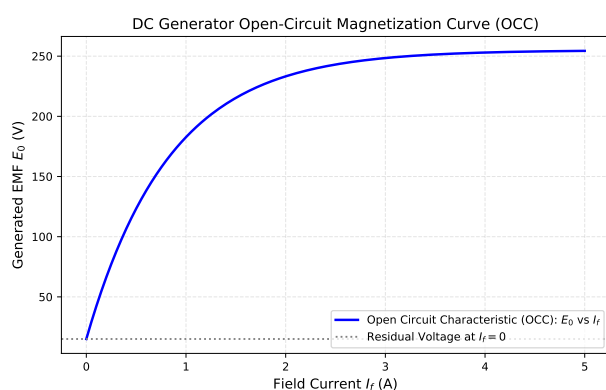
• **Commutator:**

- *In DC Generator:* Acts as a mechanical synchronous rectifier converting the internal alternating EMF (AC) induced in armature coils into unidirectional direct voltage (DC) at external terminals.
- *In DC Motor:* Reverses the armature current direction every half-rotation to maintain unidirectional continuous electromagnetic driving torque.

- **Carbon Brushes:** Provide sliding electrical contact on rotating commutator segments. Made of carbon/graphite due to high self-lubrication, negative temperature coefficient of resistance (which suppresses commutating sparks), and softer mechanical hardness than copper.

2. Characteristics of DC Generators

1. **Open-Circuit Characteristic (OCC / Magnetization Curve) [E_0 vs I_f]:** Curve between no-load generated EMF E_0 and field current I_f at rated speed. Starts at small residual voltage ($I_f = 0$) and saturates at high I_f .
2. **Internal Characteristic [E vs I_a]:** Curve between actual generated EMF on load ($E = E_0 -$ drop due to armature reaction) and armature current I_a .
3. **External Characteristic [V vs I_L]:** Curve between terminal voltage $V = E - I_a R_a$ and load current I_L .



7.3 2082 Baishakh — Question 7 [5 Marks]

Problem Statement (2082 Baishakh, Q7)

The following test results were obtained on a 20 kVA, 2200/220 V, 50 Hz, single-phase transformer:

- **O.C. test:** 220 V, 1.1 A, 125 W (On LV side)
- **S.C. test:** 52.7 V, 8.4 A, 287 W (On HV side)

Calculate the equivalent circuit parameters referred to secondary side.

Detailed Step-by-Step Solution

1. Transformation Ratio:

$$k = \frac{V_2}{V_1} = \frac{220}{2200} = 0.1 \quad (\text{LV is Secondary, HV is Primary})$$

2. Open-Circuit (O.C.) Test Calculations (Conducted on LV / Secondary Side): Given $V_{02} = 220 \text{ V}$, $I_{02} = 1.1 \text{ A}$, $W_0 = 125 \text{ W}$:

$$\cos \phi_0 = \frac{W_0}{V_{02} I_{02}} = \frac{125}{220 \times 1.1} = \frac{125}{242} \approx 0.5165$$

$$\sin \phi_0 = \sqrt{1 - (0.5165)^2} \approx 0.8563$$

$$I_w = I_{02} \cos \phi_0 = 1.1 \times 0.5165 = 0.5682 \text{ A}$$

$$I_m = I_{02} \sin \phi_0 = 1.1 \times 0.8563 = 0.9419 \text{ A}$$

Shunt branch parameters referred to secondary:

$$R_{c2} = \frac{V_{02}}{I_w} = \frac{220}{0.5682} \approx 387.19 \Omega \quad \left[\text{or } \frac{V_{02}^2}{W_0} = \frac{220^2}{125} = 387.20 \Omega \right] \quad (65)$$

$$X_{m2} = \frac{V_{02}}{I_m} = \frac{220}{0.9419} \approx 233.57 \Omega \quad (66)$$

3. Short-Circuit (S.C.) Test Calculations (Conducted on HV / Primary Side): Given $V_{sc1} = 52.7 \text{ V}$, $I_{sc1} = 8.4 \text{ A}$, $W_{sc} = 287 \text{ W}$:

$$R_{e1} = \frac{W_{sc}}{I_{sc1}^2} = \frac{287}{(8.4)^2} = \frac{287}{70.56} \approx 4.0675 \Omega$$

$$Z_{e1} = \frac{V_{sc1}}{I_{sc1}} = \frac{52.7}{8.4} \approx 6.2738 \Omega$$

$$X_{e1} = \sqrt{Z_{e1}^2 - R_{e1}^2} = \sqrt{(6.2738)^2 - (4.0675)^2} = \sqrt{39.3607 - 16.5446} = \sqrt{22.8161} \approx 4.7766 \Omega$$

Referring equivalent series parameters to secondary (LV side) via $k^2 = (0.1)^2 = 0.01$:

$$R_{e2} = k^2 R_{e1} = 0.01 \times 4.0675 = 0.04068 \Omega \approx 0.0407 \Omega \quad (67)$$

$$X_{e2} = k^2 X_{e1} = 0.01 \times 4.7766 = 0.04777 \Omega \approx 0.0478 \Omega \quad (68)$$

$$Z_{e2} = k^2 Z_{e1} = 0.01 \times 6.2738 = 0.06274 \Omega \approx 0.0627 \Omega \quad (69)$$

Verified Final Result

Equivalent Circuit Parameters Referred to Secondary (LV Side):

$$R_{c2} = 387.20 \Omega, \quad X_{m2} = 233.57 \Omega, \quad R_{e2} = 0.0407 \Omega, \quad X_{e2} = 0.0478 \Omega$$

8 Question 8: DC Machines & Magnetic Circuits

8.1 2083 Baishakh — Question 8 [5 Marks]

Problem Statement (2083 Baishakh, Q8)

A shunt generator supplies 96 A at a terminal voltage of 200 V. The armature and shunt field resistance are $0.1\ \Omega$ and $50\ \Omega$ respectively. The iron and frictional losses are 2500 W. Find: (a) EMF generated (b) Commercial efficiency.

Step-by-Step Solution

1. Current Calculations:

- Shunt field current: $I_{sh} = \frac{V}{R_{sh}} = \frac{200}{50} = 4\text{ A}$.
- Armature current: $I_a = I_L + I_{sh} = 96 + 4 = 100\text{ A}$.

2. Generated EMF (E_g):

$$E_g = V + I_a R_a = 200 + (100 \times 0.1) = 200 + 10 = 210\text{ V} \quad (70)$$

3. Power & Efficiency Calculations:

- Electrical output power: $P_{\text{out}} = V \cdot I_L = 200 \times 96 = 19,200\text{ W} = 19.2\text{ kW}$.
- Armature copper loss: $I_a^2 R_a = (100)^2 \times 0.1 = 1000\text{ W}$.
- Shunt copper loss: $I_{sh}^2 R_{sh} = (4)^2 \times 50 = 800\text{ W}$.
- Total copper loss: $P_{Cu} = 1000 + 800 = 1800\text{ W}$.
- Stray losses (Iron + Friction): $P_{\text{stray}} = 2500\text{ W}$.
- Total losses: $P_{\text{loss}} = 1800 + 2500 = 4300\text{ W}$.
- Total mechanical input power: $P_{\text{in}} = P_{\text{out}} + P_{\text{loss}} = 19,200 + 4300 = 23,500\text{ W}$.

$$\eta_c = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% = \frac{19,200}{23,500} \times 100\% \approx 81.70\% \quad (71)$$

Verified Final Result

(a) Generated EMF $E_g = 210\text{ V}$, (b) Commercial Efficiency $\eta_c = 81.70\%$

8.2 2082 Bhadra — Question 8 [5 Marks]

Problem Statement (2082 Bhadra, Q8)

A circular ring of magnetic material has a mean length of 0.5 m, cross sectional area of 100 cm^2 and an air gap of 4 mm. Calculate the MMF required to produce flux of 2 mWb in the air gap, if ring is wound with a coil of 200 turns. Take relative permeability of 800.

Step-by-Step Solution

1. Given Data: $l_i = 0.5 \text{ m}$ (iron path), $l_g = 4 \text{ mm} = 4 \times 10^{-3} \text{ m}$, $A = 100 \text{ cm}^2 = 0.01 \text{ m}^2$, $\Phi = 2 \text{ mWb} = 2 \times 10^{-3} \text{ Wb}$, $\mu_r = 800$, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$.

2. Magnetic Reluctance Method:

$$B = \frac{\Phi}{A} = \frac{2 \times 10^{-3}}{0.01} = 0.2 \text{ T}$$

$$S_i = \frac{l_i}{\mu_0 \mu_r A} = \frac{0.5}{(4\pi \times 10^{-7})(800)(0.01)} = \frac{0.5}{1.0053 \times 10^{-5}} \approx 49,735.9 \text{ AT/Wb}$$

$$S_g = \frac{l_g}{\mu_0 A} = \frac{4 \times 10^{-3}}{(4\pi \times 10^{-7})(0.01)} = \frac{4 \times 10^{-3}}{1.2566 \times 10^{-8}} \approx 318,309.9 \text{ AT/Wb}$$

$$S_{\text{total}} = S_i + S_g = 49,735.9 + 318,309.9 = 368,045.8 \text{ AT/Wb}$$

3. Total Required MMF:

$$\text{MMF} = \Phi \times S_{\text{total}} = (2 \times 10^{-3}) \times 368,045.8 \approx 736.09 \text{ AT} \quad (72)$$

Note: If net iron path is taken as $l'_i = 0.5 - 0.004 = 0.496 \text{ m}$, $\text{MMF} = 735.30 \text{ AT}$. Exciting current $I = \frac{736.09}{200} \approx 3.68 \text{ A}$.

Verified Final Result

Total MMF Required = 736.09 AT (Exciting current $I = 3.68 \text{ A}$)

8.3 2082 Baishakh — Question 8 [4 + 2 = 6 Marks]

Problem Statement (2082 Baishakh, Q8)

Explain the production of torque in DC motor with necessary mathematical explanation and derivation. What is the role of back emf in DC motor?

Part A: Derivation of DC Motor Torque Equation

When armature conductors carrying current $I_c = \frac{I_a}{A}$ lie within magnetic field B , each conductor experiences Lorentz force $F = BI_c l$.

- Magnetic flux density: $B = \frac{\text{Total Flux}}{\text{Cylindrical Area}} = \frac{P\Phi}{2\pi r l}$.
- Torque per conductor: $T_c = F \cdot r = (BI_c l)r = \left(\frac{P\Phi}{2\pi r l}\right) \left(\frac{I_a}{A}\right) l r = \frac{P\Phi I_a}{2\pi A}$.
- Total electromagnetic torque T_a for all Z armature conductors:

$$T_a = Z \cdot T_c = \frac{PZ\Phi I_a}{2\pi A} = 0.159 \left(\frac{PZ}{A}\right) \Phi I_a \text{ N} \cdot \text{m} \quad (73)$$

Part B: Role and Significance of Back EMF (E_b)

1. **Self-Regulating Governor:** Armature current is governed by $I_a = \frac{V - E_b}{R_a}$. When mechanical shaft load increases, speed N momentarily falls $\Rightarrow E_b \propto N\Phi$ decreases $\Rightarrow I_a$ rises sharply $\Rightarrow$ developed torque increases to match load demand automatically.
2. **Electromechanical Energy Conversion:** Multiplying voltage balance $V = E_b + I_a R_a$ by I_a :

$$VI_a = E_b I_a + I_a^2 R_a$$

VI_a is electrical power input, $I_a^2 R_a$ is heat loss, and $E_b I_a = P_{\text{mech}}$ represents the converted mechanical power.

8.4 2081 Ashwin — Question 8 [6 Marks]

Problem Statement (2081 Ashwin, Q8)

A 240 V dc shunt motor has armature winding resistance of $0.4\ \Omega$ and field winding resistance of $120\ \Omega$. It draws a current of 30 A at half load and the corresponding speed is 1400 rpm. If a resistance of $1.2\ \Omega$ is connected in series with the armature winding and load torque is decreased by 20%, calculate the new speed.

Step-by-Step Solution

1. Initial Condition 1 (Half Load): $V = 240\text{ V}$, $I_{L1} = 30\text{ A}$, $N_1 = 1400\text{ rpm}$, $R_a = 0.4\ \Omega$, $R_{sh} = 120\ \Omega$:

$$I_{sh} = \frac{V}{R_{sh}} = \frac{240}{120} = 2\text{ A} \quad (\text{constant flux } \Phi)$$

$$I_{a1} = I_{L1} - I_{sh} = 30 - 2 = 28\text{ A}$$

$$E_{b1} = V - I_{a1}R_a = 240 - 28(0.4) = 240 - 11.2 = 228.8\text{ V}$$

2. Modified Condition 2 (Armature Resistance Inserted & Torque Reduced):

- Series resistance added: $R_{\text{ext}} = 1.2\ \Omega \implies R_{a2} = 0.4 + 1.2 = 1.6\ \Omega$.
- Load torque reduced by 20%: $T_2 = 0.8T_1$.
- Since flux Φ is constant, torque $T \propto I_a$:

$$\frac{T_2}{T_1} = \frac{I_{a2}}{I_{a1}} \implies 0.8 = \frac{I_{a2}}{28} \implies I_{a2} = 0.8 \times 28 = 22.4\text{ A}$$

- New back EMF E_{b2} :

$$E_{b2} = V - I_{a2}R_{a2} = 240 - 22.4(1.6) = 240 - 35.84 = 204.16\text{ V}$$

3. Calculation of New Speed N_2 : Since $E_b \propto N\Phi$ (with $\Phi = \text{constant}$):

$$\frac{E_{b2}}{E_{b1}} = \frac{N_2}{N_1} \implies N_2 = N_1 \left(\frac{E_{b2}}{E_{b1}} \right) = 1400 \times \left(\frac{204.16}{228.8} \right) \approx 1249.25\text{ rpm} \quad (74)$$

Verified Final Result

New Operating Motor Speed $N_2 = 1249.25\text{ rpm}$

9 Question 9: Three-Phase Induction Motors & Transformer Efficiency

9.1 2083 Baishakh — Question 9 [5 Marks]

Problem Statement (2083 Baishakh, Q9)

A three-phase induction motor is wound for 4 poles and supplied from 50 Hz supply system. Calculate: (a) The speed of motor when the slip is 5% (b) The rotor emf frequency when the motor runs at 600 rpm.

Step-by-Step Solution

1. Synchronous Speed (N_s):

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

2. Part (a): Speed at Slip $s = 0.05$:

$$N = N_s(1 - s) = 1500(1 - 0.05) = 1500 \times 0.95 = 1425 \text{ rpm}$$

3. Part (b): Rotor EMF Frequency at $N = 600$ rpm:

$$s = \frac{N_s - N}{N_s} = \frac{1500 - 600}{1500} = \frac{900}{1500} = 0.6 \quad (60\%)$$

$$f_r = s \cdot f = 0.6 \times 50 = 30 \text{ Hz}$$

Verified Final Result

(a) Motor Speed $N = 1425$ rpm, (b) Rotor EMF Frequency $f_r = 30$ Hz

9.2 2082 Bhadra — Question 9 [5 Marks]

Problem Statement (2082 Bhadra, Q9)

A 150 kVA single phase transformer has an iron loss of 700 W and a full load copper loss of 1800 W. Calculate the copper losses, iron losses, output power and efficiency of transformer at 0.8 power factor lagging when secondary is 25% overloaded.

Step-by-Step Solution

1. Loading Parameters: $S = 150 \text{ kVA}$, $\cos \phi = 0.8 \text{ lag}$, $P_i = 700 \text{ W} = 0.7 \text{ kW}$, $P_{Cu,FL} = 1800 \text{ W} = 1.8 \text{ kW}$. Loading factor for 25% overload: $x = 1 + 0.25 = 1.25$.

2. Losses and Power:

- Iron Loss (independent of load): $P_i = 700 \text{ W} = 0.7 \text{ kW}$.
- Copper Loss at 25% overload:

$$P_{Cu,x} = x^2 P_{Cu,FL} = (1.25)^2 \times 1800 = 1.5625 \times 1800 = 2812.5 \text{ W} = 2.8125 \text{ kW}$$

- Output Active Power:

$$P_{\text{out}} = x \cdot S \cdot \cos \phi = 1.25 \times 150 \times 0.8 = 150 \text{ kW}$$

- Total Losses: $P_{\text{loss}} = P_i + P_{Cu,x} = 0.7 + 2.8125 = 3.5125 \text{ kW}$.

3. Efficiency Calculation:

$$\eta = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{loss}}} \times 100\% = \frac{150}{150 + 3.5125} \times 100\% = \frac{150}{153.5125} \times 100\% \approx 97.71\% \quad (75)$$

Verified Final Result

$$P_i = 700 \text{ W}, \quad P_{Cu} = 2812.5 \text{ W}, \quad P_{\text{out}} = 150 \text{ kW}, \quad \eta = 97.71\%$$

9.3 2082 Baishakh — Question 9 [6 Marks]

Problem Statement (2082 Baishakh, Q9)

How does three phase induction motor start? Why speed of this motor never reaches synchronous speed?

1. Starting and Operating Mechanism of 3-Phase Induction Motor

- Production of Rotating Magnetic Field (RMF):** When balanced 3-phase currents displaced by 120° electrical flow through stator windings spaced 120° apart in space, a constant-magnitude magnetic field ($\Phi_R = 1.5\Phi_m$) rotating at synchronous speed $N_s = \frac{120f}{P}$ is generated in the air gap.
- Rotor EMF Induction:** The RMF sweeps across the stationary rotor conductors at relative speed N_s , inducing alternating EMFs by Faraday's Law.
- Torque Production:** Because rotor bars form a closed electrical circuit (via end rings), the induced EMF circulates large rotor currents. These currents interact with the stator magnetic field to create Lorentz driving forces ($F = BIL$), accelerating the rotor in the direction of the RMF (per Lenz's Law).

2. Why Motor Speed Can Never Reach Synchronous Speed ($N < N_s$)

- Induction motor torque requires **relative motion** ($N_s - N$) between the stator RMF and the rotor conductors.
- If the rotor reached synchronous speed ($N = N_s$):
 - Relative speed would become zero: $N_s - N = 0$.
 - Rate of magnetic flux cutting would drop to zero $\implies$ Induced rotor EMF $e_r = 0$.
 - Rotor current $i_r = 0 \implies$ Electromagnetic torque $T = 0$.
- With zero driving torque, mechanical friction and windage immediately decelerate the rotor, ensuring it always operates with a finite positive slip $s = \frac{N_s - N}{N_s} > 0$.

9.4 2081 Ashwin — Question 9 [6 Marks]

Problem Statement (2081 Ashwin, Q9)

Explain the Torque-speed characteristics of three phase induction motor with the help of proper mathematics and graph.

Mathematical Derivation of Torque Equation

Per-phase rotor current at slip s : $I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}}$, with power factor $\cos \phi_2 = \frac{R_2}{\sqrt{R_2^2 + (sX_2)^2}}$. The developed electromagnetic torque is:

$$T = \frac{3}{2\pi N_s/60} \cdot I_2^2 \left(\frac{R_2}{s} \right) = \frac{ksE_2^2 R_2}{R_2^2 + (sX_2)^2} \quad (76)$$

Operating Regions & Characteristic Curve

1. **Low-Slip Stable Region** ($s \approx 0$, near N_s): Since s is very small, $(sX_2)^2 \ll R_2^2$:

$$T \approx \left(\frac{kE_2^2}{R_2} \right) s \Rightarrow T \propto s \propto (N_s - N) \quad (\text{Linear operating zone})$$

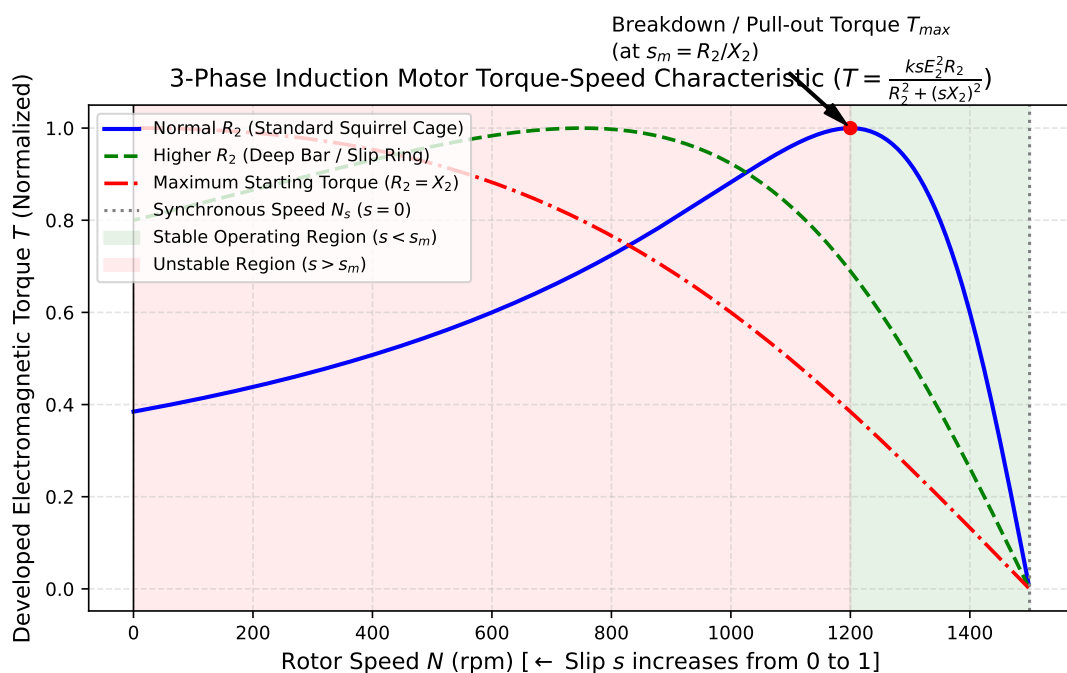
2. **High-Slip Unstable Region** ($s \approx 1$, near standstill): Since s is large, $(sX_2)^2 \gg R_2^2$:

$$T \approx \left(\frac{kE_2^2 R_2}{X_2^2} \right) \frac{1}{s} \Rightarrow T \propto \frac{1}{s} \quad (\text{Rectangular hyperbola})$$

3. **Maximum / Breakdown Torque** ($T_{\max}$): Differentiating T with respect to s yields maximum torque at slip:

$$s_m = \frac{R_2}{X_2} \Rightarrow T_{\max} = \frac{kE_2^2}{2X_2} \quad (77)$$

$T_{\max}$ is independent of rotor resistance R_2 , but adding external rotor resistance shifts s_m towards unity, allowing maximum torque at startup ($R_2 = X_2$).



10 Question 10: Conceptual Justifications & Short Notes

10.1 2083 Baishakh — Question 10 [$3 \times 3 = 9$ Marks]

(a) Series and Parallel Magnetic Circuits

- **Series Magnetic Circuit:** A magnetic circuit where the same magnetic flux Φ passes through all sections in series (e.g., iron core and air gap). Total reluctance is the algebraic sum: $S_{\text{total}} = S_1 + S_2 + \cdots + S_n$. Total MMF required: $\text{MMF} = \Phi S_{\text{total}} = H_1 l_1 + H_2 l_2 + \cdots + H_n l_n$.
- **Parallel Magnetic Circuit:** A magnetic circuit with two or more flux paths. Total flux divides across parallel branches ($\Phi = \Phi_1 + \Phi_2$). The MMF across each parallel branch is identical ($\text{MMF} = \Phi_1 S_1 = \Phi_2 S_2$).

(b) Universal Motor

A series-wound commutator motor capable of operating on either single-phase AC or DC supply.

- **Principle:** Under AC supply, both armature current I_a and field flux Φ reverse simultaneously every half-cycle, maintaining positive unidirectional torque ($T \propto \Phi I_a > 0$).
- **Design Features:** Stator and rotor iron are completely laminated to reduce eddy current losses.
- **Applications:** High-speed appliances including vacuum cleaners, blenders, food processors, and portable power drills.

(c) DC Motor Starter

- **Need for Starter:** At standstill ($N = 0$), back EMF $E_b = 0$. Since armature resistance R_a is very small ($< 1 \Omega$), direct online starting draws a destructive current $I_{a,\text{start}} = \frac{V}{R_a}$ (10 to 20 times rated current), which would burn out windings and blow fuses.
- **Function:** A starter (e.g., 3-point or 4-point starter) inserts a variable external resistance in series with the armature during startup, cutting it out progressively as back EMF E_b builds up.

(d) Voltage Regulation of Transformer

- **Definition:** The fractional or percentage change in secondary terminal voltage from no-load (V_{20}) to full-load (V_2), keeping primary supply voltage constant:

$$\% \text{ Regulation} = \frac{V_{20} - V_2}{V_2} \times 100\% \approx \frac{I_2 R_{e2} \cos \phi \pm I_2 X_{e2} \sin \phi}{V_2} \times 100\% \quad (78)$$

(+ for lagging power factor, – for leading power factor).

10.2 2082 Bhadra — Question 10 [$3 \times 3 = 9$ Marks]

(a) Speed of 3-Phase Induction Motor is Always Less than Synchronous Speed

Justification: Induction motor torque generation fundamentally requires rate of flux cutting $\frac{d\Phi}{dt} \neq 0$ between stator RMF and rotor conductors. At synchronous speed $N = N_s$, relative velocity is zero $\Rightarrow$ zero induced EMF $\Rightarrow$ zero rotor current $\Rightarrow$ zero torque. Mechanical friction prevents the rotor from ever maintaining $N = N_s$.

(b) Iron Losses of Transformer are Independent of Load

Justification: Iron losses comprise hysteresis loss ($P_h \propto B_m^{1.6} f$) and eddy current loss ($P_e \propto B_m^2 f^2$). Since primary voltage $V_1 \approx 4.44 f N_1 \Phi_m$ is constant, peak core flux density B_m remains strictly constant from no-load to full-load. Hence, iron losses remain constant.

(c) DC Motor Draws High Starting Current when Switched on Directly

Justification: From $I_a = \frac{V - E_b}{R_a}$, at startup ($t = 0, N = 0$), $E_b = 0$. Since armature resistance R_a is negligible, $I_{a,\text{start}} = \frac{V}{R_a}$ reaches 500% to 2000% of rated full-load current.

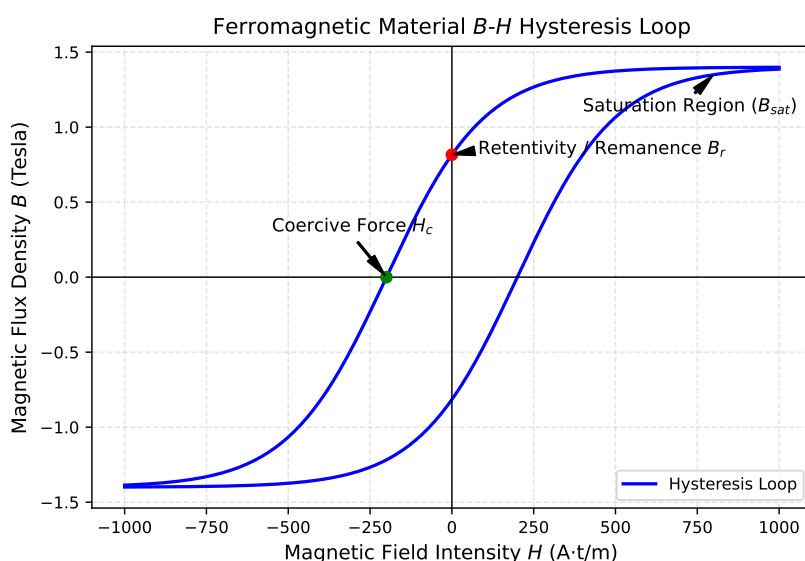
(d) Single-Phase Induction Motor Has Two Stator Windings

Justification: A single-phase stator winding produces a pulsating stationary magnetic field with zero net starting torque (by Double-Field Revolving Theory). To produce an initial rotating magnetic field, an auxiliary starting winding is placed 90° electrical apart with a series phase-splitting capacitor.

10.3 2082 Baishakh — Question 10 [$2 \times 3 = 6$ Marks]

(a) Magnetization of Magnetic Material & B - H Hysteresis Loop

Applying magnetic field intensity H aligns magnetic dipoles. The B - H curve exhibits a linear region, knee point, saturation (B_{sat}), remanence/retentivity (B_r), and coercive force (H_c). Cyclic AC magnetization causes hysteresis energy loss equal to the loop area: $P_h = \eta B_m^{1.6} f V W$.



(b) Auto-Transformer and Its Applications

A transformer with a single continuous tapped winding common to both primary and secondary.

- **Power Transfer:** Converted partly by electrical conduction and partly by magnetic induction.
- **Advantages:** Uses less copper ($\text{Weight}_{\text{auto}} = (1 - k)\text{Weight}_{2W}$), lower cost, higher efficiency, and superior voltage regulation.
- **Applications:** Laboratory Variacs, induction motor starters, transmission line booster transformers.

(c) Stepper Motor

An electromechanical digital actuator that rotates in discrete angular steps ($\theta_s = \frac{360^\circ}{mP}$) per input electrical pulse. Operates open-loop without feedback encoders in 3D printers, CNC machines, and robotics.

10.4 2081 Ashwin — Question 10 [$2 \times 3 = 6$ Marks]**(a) Role of Commutator and Carbon Brush in DC Machine**

- **Commutator:** Mechanical inverter/rectifier maintaining unidirectional torque in motors and DC output in generators.
- **Carbon Brushes:** Self-lubricating, wear-resistant sliding electrical collectors feeding current to/from rotating segments.

(b) Hysteresis and Eddy Current Power Losses

- **Hysteresis Loss** ($P_h = \eta B_m^{1.6} f V \text{ W}$): Energy lost in overcoming magnetic domain friction during cyclic AC reversal. Minimized using high-permeability silicon steel.
- **Eddy Current Loss** ($P_e = K_e B_m^2 f^2 t^2 V \text{ W}$): Ohmic $I^2 R$ power lost due to circulating currents induced in the conductive core. Minimized by laminating the core into thin, varnished sheets ($t \approx 0.35 \text{ mm}$).

(c) Stepper Motor Types and Characteristics

- **Variable Reluctance (VR):** Multi-toothed iron rotor seeking minimum reluctance path.
- **Permanent Magnet (PM):** High detent torque with permanent magnet rotor poles.
- **Hybrid Stepper Motor:** Combines VR teeth and PM core for high resolution (1.8° step angle) and high holding torque.